

1. Intellectual Property: Meaning and Importance

Intellectual Property (IP) means legal rights over creations of the mind. In technology, IP may appear as source code, software architecture, databases, ML model designs, technical inventions, brand names, logos, user interfaces, documentation, training data collections and confidential business processes. WIPO explains IP through areas such as patents, copyright, trademarks, designs, geographical indications and trade secrets.

Why IP matters for students

- **Software products are IP-heavy:** an AI app may contain copyrighted code, licensed open-source libraries, and trade-secret model tuning methods, a trademarked product name and possibly patentable technical inventions.
- **Projects often use third-party material:** GitHub code, Kaggle datasets, APIs, papers, pretrained models and images may have licenses or usage restrictions.
- **Employers care about ownership:** work created during internships or employment may be owned by the company depending on contract terms, institutional policy and law.
- **AI increases IP risk:** training data, generated code, model outputs and synthetic media can raise questions of copyright, attribution, database rights, privacy and contractual restrictions.

Type of IP	Protects	AI & ML example	Usual way of protection
Copyright	Original expression, not merely ideas	Python source code, documentation, UI artwork, datasets as compilations	Automatic protection plus licenses and contracts
Patent	Technical invention that meets legal requirements	Novel hardware accelerator or technical ML system improvement	Application and grant by patent office
Trade secret	Confidential commercially valuable information	Private training pipeline, model weights, customer analytics method	Secrecy, NDAs, access control, contracts
Trademark	Brand identifiers that distinguish goods/services	Name/logo of an AI product or startup	Use and/or registration, depending on jurisdiction

Basic principles

- IP rights are usually territorial. Protection in one country does not automatically mean identical protection everywhere.
- IP does not normally protect every idea. Many systems protect expression, inventions, brand identity or confidential information under specific conditions.
- Owning a copy is different from owning rights. Downloading code, a model or a dataset does not automatically give permission to redistribute or commercialize it.
- Good IP practice includes attribution, license checking, confidentiality discipline, documentation of authorship and ethical data use.

2. Copyrights

Copyright is the legal protection given to creators over original literary and artistic works. WIPO states that copyright covers works such as books, music, paintings, films, computer programs, databases, advertisements, maps and technical drawings. In India, the Copyright Act, 1957 is the principal legislation for copyright.

What copyright protects in computing

- **Source code and object code:** software code is usually treated as a literary work. Under the Indian Copyright Act, literary work includes computer programmes, tables and compilations including computer databases.
- **Documentation and manuals:** README files, API documentation, user guides and comments may be protected if original.
- **Databases and datasets:** a dataset may involve copyright in selection, arrangement, annotation or documentation. Individual facts are usually different from the expression or compilation.
- **Images, videos and UI assets:** icons, graphics, diagrams and interface artwork may be protected.
- **AI-generated outputs:** ownership and protectability can be complex. Students should check local law, institutional rules and platform terms before claiming ownership or reusing outputs commercially.

What copyright generally does not protect

- Bare ideas, concepts or algorithms as abstract ideas.
- Mathematical facts, formulas and general methods by themselves.
- Independent work created without copying protected expression.
- Short names or brand identifiers, which are usually handled under trademark law rather than copyright law.

Rights of a copyright owner

The exact rights depend on national law and the type of work. Common rights include reproduction, adaptation, and distribution, communication to the public and licensing. For software, this means that copying, modifying, distributing or integrating code may require permission unless a license or legal exception allows it.

Fair dealing, exceptions and software

Many copyright laws provide limited exceptions. In India, Section 52 of the Copyright Act lists certain acts that do not constitute infringement. For computer programs, there are specific provisions for lawful possessors, such as making copies or adaptations in limited circumstances, including for the purpose for which the program was supplied and for interoperability-related purposes under statutory conditions.

Some examples

- **Using a GitHub repository:** You may view public code, but reuse depends on the repository license. No license usually means no clear permission to copy, modify or redistribute.
- **Copying figures from a paper:** A citation is ethically necessary, but citation alone may not always be legal permission for reuse in a commercial product.
- **Using a dataset:** Read the dataset license carefully. A dataset may allow academic research but restrict commercial use.
- **Training with scraped data:** Copyright, terms of service, privacy and database issues may arise. Do not assume that public access equals free legal use.

A patent is an exclusive right granted for an invention. WIPO explains that patents protect inventions while also making technical information about the invention publicly available. In India, patents are governed by the Patents Act, 1970.

Patent basics

- **Invention:** A product or process that provides a new technical solution to a problem.
- **Exclusive right:** A granted patent generally lets the patent owner control how the invention is used by others in that jurisdiction.
- **Disclosure bargain:** The inventor receives a limited exclusive right in exchange for disclosing the invention to the public.
- **Term:** Under Indian patent law, the term of a patent is governed by Section 53 of the Patents Act, 1970. Students should verify the latest statutory text for exact application and conditions.

Patentability requirements in simple language

- **Novelty:** The invention should be new compared with prior knowledge.
- **Inventive step / non-obviousness:** It should not be an obvious improvement to a skilled person.
- **Industrial applicability / usefulness:** It should be capable of practical application.
- **Patentable subject matter:** The invention must not fall into excluded categories under the relevant law.

Software, algorithms and AI in India

For students in India, Section 3(k) of the Patents Act is very important. It excludes a mathematical method, business method, computer programme per se or algorithms from patentability. WIPO also describes India as treating mathematical methods, business methods, computer programmes per se and algorithms as not patentable under this provision.

This does not mean that every computer-related invention is automatically impossible to protect. However, students should not assume that an AI model, Python script or algorithm alone is patentable. Patentability depends on the complete claim, technical contribution and applicable examination practice. For real applications, professional advice is essential.

Example	Likely IP issue	Comment
A new sorting algorithm described only as mathematics	Patent exclusion risk	May be treated as algorithm/mathematical method.
A trained ML model file kept private	Trade secret / contract	May be protected by secrecy if commercially valuable and access is controlled.
A novel AI chip architecture	Patent possibility	May involve technical hardware invention if requirements are met.
A brand name for an AI chatbot	Trademark	Not a patent issue; consider trademark clearance and registration.

4. Trade Secrets and Trade Secret Laws

A trade secret is confidential information that has commercial value because it is secret and is protected through reasonable secrecy measures. WIPO states that trade secrets are IP rights on confidential information which may be sold or licensed, and that qualifying information should be commercially valuable because secret, known only to a limited group, and subject to reasonable steps to keep it secret.

Element	Meaning	Technology example
Secret	Not generally known or easily accessible to relevant circles	Internal fraud-detection rules not published publicly
Commercial value	Valuable because competitors do not know it	Private data-cleaning pipeline that improves model accuracy
Reasonable steps	Owner actively protects secrecy	NDAs, access logs, encryption, restricted repositories

Trade secret protection in India

A WIPO country sheet on India states that India does not have a separate and exclusive statute for undisclosed information or trade secrets. It notes that trade secrets and confidential information can be protected to a large extent by contracts such as NDAs and confidentiality clauses in employment contracts, and also through common law principles such as breach of confidence; in some cases, criminal law or IT law remedies may be available.

Examples in AI & ML

- Private training data preprocessing rules.
- Model weights, tuning methods and evaluation pipelines kept confidential.
- Customer behavior analytics models used by an e-commerce company.
- Source code of a proprietary recommender system.
- Security monitoring rules in a cyber-defense platform.

How trade secrets are protected in practice

- **Legal measures:** NDAs, confidentiality clauses, IP assignment clauses, vendor agreements and employee policies.
- **Technical measures:** role-based access control, encryption, repository permissions, audit logs, secret management tools and device controls.
- **Organizational measures:** training, exit interviews, need-to-know access, labelling documents as confidential and incident reporting.
- **Documentation:** clear records showing what information is confidential and who had access.

Employees often receive access to confidential information because they need it for work. This creates legal and ethical duties. In India, WIPO notes that employment contracts and confidentiality clauses are important sources of trade secret protection, but the scope depends on contractual obligations and the nature of the information.

Common employee duties

- **During employment:** Use confidential information only for authorized work; do not copy or transfer confidential files to personal accounts; follow access policies.
- **After employment:** Do not disclose or use former employer trade secrets. Return company devices, documents and repository access.
- **In academic internships:** Respect project NDAs, research collaboration rules and company data policies.
- **When publishing:** Get clearance before publishing results based on company datasets, internal metrics or proprietary code.

Important distinction: skill vs secret

An employee is usually free to use general skill, knowledge and experience learned during work. But taking confidential documents, source code, private datasets, model weights, business plans or client lists may violate confidentiality or trade secret obligations. The difficult part is separating general expertise from protected confidential information.

Practical scenarios

Scenario	Risk	Good practice
Intern downloads the company training dataset to a personal drive for later use.	High risk of confidentiality breach.	Do not copy; ask written permission; use approved systems only.
Engineer remembers general debugging techniques after leaving.	Usually normal skill/experience.	Use general skill, not copied confidential code or documents.
Employee posts internal model accuracy numbers on social media.	Possible disclosure of confidential business information.	Seek approval before public disclosure.
Student uses company API keys in a public GitHub repository.	Security breach and possible trade secret/confidentiality issue.	Use secret managers; revoke exposed keys; report immediately.

6. Key Intellectual Property Issues

AI & ML projects combine multiple IP layers. A single final-year AI project may use open-source libraries, pretrained models, datasets, research papers, a web interface, a database, a logo and a deployment platform. Each layer may have a separate owner, license or legal restriction.

Layer in AI system	Possible IP/legal issue	Student action
Source code	Copyright and open-source license compliance	Keep license files, attribution and dependency list.
Dataset	Dataset license, copyright in compilation, privacy, terms of service	Read dataset terms; do not scrape prohibited content.
Model architecture	Patentability or research publication rights	Check novelty and institutional policy.
Model weights	Trade secret, contract, license restrictions	Do not redistribute without permission.
Generated output	Copyright, plagiarism, attribution, deepfake/misuse issues	Review output; cite tools when required; avoid deceptive use.
Brand/domain	Trademark and cybersquatting risk	Search for existing marks before naming product.

7. Plagiarism

Plagiarism is an academic and professional integrity violation. UGC defines plagiarism as taking someone else's work or idea and passing them as one's own. ACM also treats plagiarism as misrepresentation of another person's writings, ideas or creative work, including documents, data, research proposals and computer code, as one's own.

Types of plagiarism relevant to students

- **Text plagiarism:** copying paragraphs from papers, websites or reports without quotation and citation.
- **Code plagiarism:** copying code from classmates, GitHub, Stack Overflow or AI tools and submitting it as original work.
- **Idea plagiarism:** using another person's research idea, experiment design or algorithmic contribution without acknowledgment.
- **Data or result plagiarism:** using another person's dataset, experimental results, graphs or tables without permission or citation.
- **Self-plagiarism:** reusing one's own earlier submitted work without disclosure where new work is required.
- **AI-assisted plagiarism:** submitting AI-generated or AI-rewritten content as if it were fully one's own when disclosure or original work is required.

UGC academic-integrity approach

UGC's 2018 regulations apply to students, faculty, researchers and staff of higher educational institutions in India and aim to create awareness, set up institutional mechanisms and prevent and punish plagiarism. The regulations include similarity levels for theses and dissertations, beginning with Level 0 for similarities up to 10% as minor similarities with no penalty, and higher levels for more serious similarity ranges.

Avoiding plagiarism in technical work

- Use quotation marks for exact copied words and cite the source.
- Paraphrase properly: rewrite in your own structure and cite the original idea.
- For code, mention the source and license in comments or documentation when reuse is allowed.
- Do not hide plagiarism by changing variable names or rearranging code.
- Keep lab notebooks and commit histories showing your own work.
- Follow your institution's AI-use policy when using ChatGPT, GitHub Copilot or other AI tools.

8. Reverse Engineering

Reverse engineering means studying a product, program, system or model to understand how it works. In software, it may involve decompilation, API analysis, file-format analysis, protocol analysis, model extraction attempts or security testing. Reverse engineering can be legal, illegal or ethically questionable depending on purpose, method, contract, copyright, trade secret, cybersecurity law and jurisdiction.

Legitimate uses

- Interoperability: making one program work with another program or file format.
- Security research: finding vulnerabilities responsibly and reporting them.
- Learning and debugging: understanding behavior of systems one is lawfully allowed to inspect.
- Data recovery and compatibility: recovering old files or ensuring compatibility with existing systems.

Legal limits and risks

- **Copyright risk:** decompiling or copying code can involve reproduction or adaptation of copyrighted software unless a legal exception or license permits it.
- **Contract risk:** EULAs, NDAs or platform terms may restrict reverse engineering even when the act is technically possible.
- **Trade secret risk:** WIPO notes that independently acquiring information through reverse engineering may not violate trade secret protection, but this depends on lawful acquisition and circumstances. The WIPO India country sheet also notes that in general reverse engineering cannot be prevented through a breach of confidentiality clause, but the complete legal position is context-sensitive.
- **Cyber law risk:** bypassing access controls, stealing credentials or attacking systems is not legitimate reverse engineering; it may become unauthorized access or computer misuse.

India and software interoperability

The Indian Copyright Act includes specific exceptions for computer programmes under Section 52, including limited acts by a lawful possessor and interoperability-related acts subject to statutory conditions. Students should read the exact statutory text before relying on an exception.

9. Open Source Code

Open source software is software distributed under licenses that allow users to use, study, modify and share the software under specified terms. OSI states that open-source licenses comply with the Open Source Definition and, in brief, allow software to be freely used, modified and shared.

Open source does not mean no rules

- A license is still legally important. It tells you what you may do and what obligations you must follow.
- Public GitHub code without a license is not automatically open source. It may be visible but not clearly reusable.
- Different licenses have different obligations: attribution, notice preservation, source-code disclosure, patent clauses or share-alike/copyleft rules.
- SPDX provides standardized short identifiers and canonical license references that help software teams track licenses.

License family	Typical idea	Practical note for students
Permissive licenses	Generally allow broad reuse with notice/attribution obligations	Examples include MIT, BSD and Apache-style licenses; still keep notices.
Copyleft licenses	May require derivative works to be distributed under the same or compatible license	Be careful when combining with proprietary code.
Source-available but not open source	Code may be visible but usage may be restricted	Do not call it open source unless it meets open-source criteria.
Dataset/model licenses	May differ from software licenses	Check commercial-use, redistribution and attribution restrictions.

10. Competitive Intelligence

Competitive intelligence (CI) means gathering and analyzing information about competitors, markets, customers and technologies to support lawful business decisions. Ethical CI uses legal and honest sources. It is different from industrial espionage, hacking, bribery, deception or stealing trade secrets.

Acceptable sources of competitive intelligence

- Public websites, product documentation, advertisements and press releases.
- Published research papers, patents and trademark databases.
- Conference talks, public demos and official product announcements.
- Lawfully obtained market reports and customer feedback.
- Job postings and publicly available technical blogs, used responsibly.

Unethical or risky methods

- Pretending to be a customer, journalist or job applicant to obtain confidential information.
- Asking employees or ex-employees to reveal trade secrets from their current or former employer.
- Hacking systems, scraping in violation of access controls or using stolen credentials.
- Collecting competitively sensitive information directly from competitors in a way that may raise competition-law concerns.
- Using confidential documents accidentally received from a competitor without reporting and legal guidance.

Competition law angle

Competitive intelligence must also respect competition law. In India, the Competition Act, 2002 prohibits anti-competitive agreements and deals with practices such as price fixing, market sharing, output control and bid rigging. Therefore, companies should be careful about direct exchanges of sensitive pricing, output, future strategy or bidding information with competitors.

Ethical CI principles

- Be truthful about identity and purpose when interacting with people.
- Use legal sources and methods.
- Respect confidentiality requests and contractual restrictions.
- Do not induce others to breach NDAs or employment duties.
- Document sources so that information can be verified.

Professional CI bodies such as SCIP provide ethical guidance for competitive and market intelligence practice.

11. Trademark Infringement

A trademark is a sign capable of distinguishing the goods or services of one enterprise from those of other enterprises. WIPO states that trademarks are protected by IP rights and that registration generally gives legal certainty and strengthens enforcement. In India, the Trade Marks Act, 1999 governs registration and protection of trademarks.

What can be a trademark?

- Brand names, product names and service names.
- Logos, symbols, labels and packaging elements.
- Slogans or distinctive phrases in some cases.
- Domain names may also function like brand identifiers when used commercially.

Trademark infringement in simple terms

Trademark infringement occurs when a person uses a mark in trade in a way that violates the rights of a registered trademark owner, often because the mark is identical or confusingly similar and likely to cause confusion, association or unfair advantage. Section 29 of the Indian Trade Marks Act deals with infringement of registered trademarks.

Situation	Risk	Example
Identical mark for identical services	High confusion risk	Launching "OpenAI Tutor" for an AI tutor without authorization.
Similar mark for similar goods/services	Confusion or association risk	Using a logo/name close to a known AI platform.
Using brand in misleading domain	Trademark and cybersquatting risk	Registering a domain suggesting official association.
Descriptive honest use	May be allowed depending on facts	Saying "compatible with Python" truthfully, without implying sponsorship.

12. Cybersquatting

Cybersquatting generally means registering, using or trafficking in a domain name that is identical or confusingly similar to someone else's trademark or name, usually in bad faith. It commonly involves registering domains to sell them to trademark owners, divert users, run phishing attacks, display ads or damage a brand.

International and Indian dispute systems

ICANN's Uniform Domain-Name Dispute-Resolution Policy (UDRP) addresses disputes arising from abusive domain-name registrations, including cybersquatting, through administrative proceedings initiated by trademark holders. For .IN and .Bharat domain names, the .IN Domain Name Dispute Resolution Policy (INDRP), adopted by NIXI, sets the framework for disputes between a domain registrant and a complainant.

Typical elements considered in domain disputes

- The domain name is identical or confusingly similar to a name, trademark or service mark in which the complainant has rights.
 - The registrant has no rights or legitimate interests in the domain name.
 - The domain name has been registered or is being used in bad faith or for an unlawful purpose.
- These elements are reflected in the .IN policy framework.

Examples

Domain Behavior	Likely issue
Registering a famous brand plus "support" and hosting a fake login page	Cybersquatting, phishing and cybercrime risk
Registering a competitor's product name to sell it back at a high price	Bad-faith domain registration risk
Using a domain for a genuine fan or criticism site without commercial confusion	May require careful legal analysis; not automatically cybersquatting
Registering a domain based on your own original startup name	Usually legitimate if no conflict with prior rights

13. Comparison Tables

A. IP Comparison Table

IP type	Protects	Does not mainly protect	Duration / condition	Common AI example
Copyright	Original expression such as code, docs, images	Bare ideas or algorithms as abstract concepts	Automatic; duration depends on law and work type	Python code and documentation
Patent	Technical inventions	Excluded subject matter, mere abstract algorithm	Limited term after grant, subject to law	AI chip or technical system improvement
Trade secret	Valuable confidential information	Information publicly known or independently developed	As long as secrecy and reasonable protection continue	Private model weights and training pipeline
Trademark	Brand identity and consumer source indication	Technical invention or code expression	Can be renewed if legal requirements are met	Name and logo of an AI product

B. Common Mistake vs Correct Practice

Mistake	Why it is risky	Correct practice
Copying code without checking license	Copyright/license violation and plagiarism risk	Check license, attribute, follow conditions
Publishing internship source code	Breach of confidentiality/trade secret risk	Get written permission or remove confidential material
Using a famous brand name for an app	Trademark confusion and infringement risk	Choose original name and run trademark/domain search
Scraping a website against its terms	Contract, copyright, privacy and cyber-law risk	Use permitted datasets/APIs and document permissions
Registering a competitor-like Domain	Cybersquatting and bad-faith risk	Use own brand and avoid confusing similarity

Quality Software

1.1 Meaning of quality software

Quality software is software that correctly performs its intended function, is reliable in real use, is secure against reasonable threats, is usable by its intended users, can be maintained over time, and satisfies the requirements agreed with stakeholders. A program may run successfully in a demo but still be poor-quality software if it fails under load, exposes private data, is impossible to maintain, or produces unsafe results.

In cyber law and ethics, software quality is important because software now affects finance, healthcare, education, transport, public services, and personal privacy. Poor quality is not only a technical issue; it can become an ethical, legal, and social issue.

1.2 Main attributes of quality software

Quality attribute	Simple meaning	Example
Functional suitability	The software provides the functions that users actually need.	A question-generation tool generates MCQs, short-answer questions, and answer keys as promised.
Reliability	The software works consistently and does not fail frequently.	An exam portal remains available during peak submission time.
Performance efficiency	The software uses time, memory, CPU, GPU, and network resources efficiently.	A chatbot responds within acceptable time even when many students use it.
Usability	Users can understand and use the system without unnecessary difficulty.	A teacher can upload a PDF and generate questions without knowing command-line tools.
Security	The system protects data and prevents unauthorized access or misuse.	Student marks and personal data are encrypted and access-controlled.
Compatibility	The software works with other systems and environments as required.	An AI model API integrates with a learning management system.
Maintainability	The software can be corrected, improved, tested, and adapted easily.	Developers can update a model, retrain it, and test it without breaking the application.
Portability	The software can run in different environments with reasonable effort.	A Python ML application can be deployed locally, on cloud, and in Docker.

The above list follows the widely used structure of the ISO/IEC 25010 software product quality model. Some newer sources discuss AI-specific quality models as extensions because AI systems may learn from data, behave probabilistically, and change performance after deployment.

1.3 Quality in software is more than accuracy

In many projects, students focus only on model accuracy. Accuracy is important, but it is not enough. A high-accuracy model may still be low-quality if it is biased, insecure, not explainable, slow, unstable under new data, or trained on poor-quality data.

Quality concern	Why it matters	Example
Data quality	The model learns from data; wrong or biased data can create wrong or unfair outputs.	A hiring model trained on biased historical data may disadvantage certain applicants.
Generalization	The model should work on new, unseen data, not only on training data.	A medical image model trained on one hospital dataset may fail on images from another hospital.
Robustness	Small changes or attacks should not easily break the system.	A stop-sign recognition model should not fail because of small stickers or lighting changes.
Explainability	Users and auditors may need understandable reasons for high-impact decisions.	A loan rejection system should provide meaningful reasons, not only a score.
Monitoring and drift control	Model performance may degrade when real-world data changes.	A fraud-detection model may become less useful as attackers change their methods.
Human oversight	Human review is necessary for high-risk decisions.	An AI diagnosis support tool should assist doctors, not silently replace clinical judgment.

1.4 Difference between quality assurance, quality control, testing, and verification

Term	Simple meaning
Quality assurance (QA)	Process-focused activities that help prevent defects before or during development.
Quality control (QC)	Product-focused checks that detect defects in the finished or intermediate product.
Software testing	Execution or evaluation of software to find defects and check expected behavior.
Verification	Checking whether the product is being built according to specified requirements and design.
Validation	Checking whether the right product is being built for real user needs.

1.5 Simple example: low-quality versus quality software

Suppose a team builds an AI-based attendance system using face recognition.

Low-quality version	Quality-focused version
Works only in a classroom demo.	Tested under different lighting, camera angles, and student groups.
Stores face data without clear security controls.	Encrypts sensitive data and applies access control.
No process for correcting false attendance.	Allows teacher review and correction with audit logs.
No bias or fairness evaluation.	Measures error rates across relevant groups and improves data collection.
No documentation.	Maintains requirements, design, test reports, risk register, and user guide.

Importance of Software Quality

2.1 Why software quality is important

Software controls banking transactions, hospital records, traffic systems, online classes, e-commerce payments, smartphones, cloud services, and AI applications. When software quality is poor, the result may be inconvenience, financial loss, privacy violation, safety risk, reputational damage, or legal dispute.

Area affected	How poor quality causes harm	Example
Users	Loss of time, money, privacy, or trust.	A buggy payment app deducts money but fails to confirm the transaction.
Business	Higher maintenance cost, customer complaints, lost contracts, and brand damage.	A SaaS product loses customers because of frequent outages.
Security	Defects become vulnerabilities.	An input validation flaw leads to SQL injection.
Law and regulation	Failure to meet legal duties, contract promises, or sector rules.	A health app leaks sensitive records because of weak access control.
Society	Large-scale systems can affect public welfare.	A public exam portal fails during a national-level examination.
AI ethics	Biased or unreliable AI can create unfair outcomes.	An automated scoring model disadvantages certain student groups.

2.2 Cost of poor quality

The later a defect is found, the more expensive it usually becomes to fix. A requirements mistake found during planning may require a small correction. The same mistake found after deployment may require code changes, database migration, user communication, compensation, and legal review.

Type of cost	Meaning	Example
Prevention cost	Cost spent to avoid defects.	Training, code review, secure design, test planning.
Appraisal cost	Cost spent to find defects.	Testing, inspection, static analysis, audits.
Internal failure cost	Cost of defects found before release.	Rework, debugging, retesting.
External failure cost	Cost of defects found after release.	Support calls, patches, refunds, breach response, lawsuits.

2.3 Ethical importance

Professional ethics requires developers and organizations to avoid foreseeable harm. A developer who knowingly ignores security flaws, false results, or safety risks is not merely making a technical mistake; the developer may be acting irresponsibly toward users and society. For students, ethical software quality includes fairness, transparency, privacy protection, and accountability.

2.4 Importance in cyber law

Cyber law deals with digital rights, responsibilities, evidence, contracts, offences, data protection, and liability. Software quality connects to cyber law because insecure or defective software can cause unauthorized access, data leakage, service failure, incorrect records, or harm to consumers. Good documentation, testing, audit logs, and change control can also become evidence that the organization acted responsibly.

Software Product Liability

3.1 Simple meaning of product liability

Product liability means legal responsibility for harm caused by a defective product or deficient service. In software, liability questions arise when software fails, produces wrong output, creates security weaknesses, violates privacy, or causes physical, financial, or other harm. Liability depends on the country, contract terms, product type, industry, and facts of the case.

Software product liability is complex because software may be sold as a product, delivered as a service, updated continuously, integrated with hardware, or embedded inside an AI system. Therefore, students should not assume that software is legally harmless merely because it is intangible code.

3.2 Common bases of liability in software projects

Basis	Simple meaning	Software example
Contractual liability	Failure to meet promises in a contract, service-level agreement, or specification.	A vendor promises 99.9% uptime but repeatedly fails without valid exception.
Negligence	Failure to take reasonable care, causing foreseeable harm.	A team ignores known security vulnerabilities in a payment system.
Product defect	The product is defective in design, manufacture, warning, documentation, or safety.	An insulin-pump software update causes unsafe dosage calculation.
Deficient service	The service is not provided with reasonable care or expected quality.	A cloud provider loses customer data because backups were not maintained as promised.
Data/privacy breach	Poor software controls expose personal or sensitive data.	A school app leaks student records due to weak authentication.
Misrepresentation	Marketing claims or documentation overstate capabilities.	An AI tool is advertised as fully accurate for diagnosis though it has not been clinically validated.

3.3 Types of software defects relevant to liability

Defect type	Meaning	Example
Requirements defect	The required behavior was misunderstood or incomplete.	The system has no consent mechanism for collecting user data.
Design defect	The architecture or algorithm creates risk.	A safety system has no fail-safe mode when the sensor fails.
Coding defect	Programming error causes wrong behavior.	Integer overflow causes incorrect billing or dosage calculation.
Testing defect	Insufficient testing fails to detect foreseeable problems.	No test for large file upload, causing server crash during exams.
Documentation or warning defect	Users are not informed of limitations or safe use.	AI chatbot does not warn that outputs may be wrong and need human verification.
Maintenance/update defect	A patch or update introduces new risks.	An update disables important security settings.

3.4 Liability risk in AI/ML systems

AI systems create special risk because outputs are often probabilistic, data-dependent, and difficult to explain. A traditional rule-based program normally follows explicit instructions. An ML model learns patterns from data and may behave incorrectly on new data, biased data, or adversarial inputs.

AI/ML risk	Why it can create liability concern	Control measure
False prediction	Wrong decision may harm a person or organization.	Human review, confidence thresholds, validation on representative data.
Bias and unfairness	Discriminatory outcomes may violate ethical or legal expectations.	Bias testing, diverse data, fairness metrics, impact assessment.
Lack of explainability	Affected persons and auditors may not understand the basis of a decision.	Use interpretable models where possible, explanations, documentation.
Data drift	Performance may degrade after deployment.	Continuous monitoring, drift detection, periodic retraining.
Unsafe automation	System may take action without adequate human control.	Human-in-the-loop for high-risk decisions, safe fallback.
Security attacks	Poisoning, evasion, prompt injection, or model extraction may cause harm.	Threat modeling, red teaming, access control, secure pipeline.

3.5 Defensive practices: how teams reduce liability risk

- Write clear requirements and acceptance criteria. Ambiguity creates disputes.
- Maintain traceability from requirements to design, code, tests, risks, and release decisions.
- Use code review, testing, static analysis, security testing, and user acceptance testing.
- Document known limitations. Do not hide material risks from users.
- Use change management and release notes for every update.
- Maintain logs, audit trails, incident response plans, and rollback plans.
- For AI systems, document datasets, model versions, performance metrics, bias tests, monitoring plans, and human oversight design.
- Follow applicable standards, policies, and sector-specific regulations.

Software Development Process

4.1 Meaning of software development process

A software development process is a structured way to plan, build, test, release, operate, and maintain software. A process does not guarantee perfect software, but it reduces confusion, improves communication, creates evidence of due care, and helps teams detect risks earlier.

4.2 Basic SDLC phases

Phase	Main work	Quality focus
Planning	Define goals, scope, feasibility, team, schedule, budget, risk.	Avoid unrealistic promises and unmanaged risk.
Requirements analysis	Understand what users and stakeholders need.	Clear, testable, complete requirements.
Design	Choose architecture, data flow, UI, database, APIs, security controls.	Design for reliability, maintainability, security, and scalability.
Implementation	Write code, configure infrastructure, integrate components.	Coding standards, code review, secure coding.
Testing and verification	Check that software meets requirements and quality expectations.	Unit, integration, system, security, performance, and acceptance testing.
Deployment	Release to users, migrate data, configure production environment.	Rollback plan, monitoring, release documentation.
Operation and maintenance	Fix defects, improve features, monitor performance and security.	Change control, patches, incident response, user support.
Retirement	Safely remove or replace the system.	Data migration, archival, legal retention, user communication.

4.3 Common process models

Model	Simple explanation	When useful	Quality risk if misused
Waterfall	Sequential phases: requirements, design, coding, testing, release.	Stable requirements, regulated documentation-heavy projects.	Late discovery of requirement mistakes.
V-Model	Testing is planned in parallel with each development phase.	Safety, defense, medical, or high-assurance systems.	Can be rigid if requirements change rapidly.
Iterative/incremental	Build the product in repeated cycles and add features gradually.	Complex systems where feedback is needed.	Poor architecture may accumulate technical debt.
Agile/Scrum	Short sprints, frequent feedback, working software increments.	Dynamic projects and user-centered development.	May ignore documentation or long-term quality if badly managed.
DevOps	Integrates development and operations with automation and monitoring.	Cloud services and frequent releases.	Fast release without proper testing can spread defects quickly.
Secure SDLC	Security is included from requirements to maintenance.	Systems handling Personal data, money, or critical services.	Security added only at the end becomes expensive and incomplete.

4.4 Secure software development process

In a cyber law and ethics course, secure development is essential. Security should not be treated as a final checklist. It should be included from the beginning through threat modeling, secure design, dependency management, code review, testing, vulnerability handling, and incident response.

Secure SDLC activity	Explanation
Threat modeling	Think like an attacker and identify what can go wrong.
Secure requirements	Write security requirements such as authentication, authorization, encryption, logging, and privacy controls.
Secure coding	Avoid common mistakes such as injection, hard-coded secrets, weak randomness, and unsafe deserialization.
Dependency management	Track third-party packages and update vulnerable libraries.
Static and dynamic testing	Use tools and manual review to find vulnerabilities in code and running systems.
Release security review	Check whether known critical risks are addressed before deployment.
Incident response	Plan how to respond if a vulnerability or breach is found.

4.5 AI/ML software development lifecycle

AI/ML development includes the normal software lifecycle plus data and model lifecycle activities. Teams must manage data collection, labeling, preprocessing, training, evaluation, deployment, monitoring, retraining, and model retirement.

4.6 Documentation produced during development

- Requirements specification and user stories.
- Design document, architecture diagram, and data-flow diagram.
- Risk register and threat model.
- Software quality assurance plan and test plan.
- Test cases, test results, defect reports, and acceptance report.
- Source-code repository history, code review records, and release notes.
- User manual, admin manual, privacy/security notice, and known limitations.
- For AI/ML: dataset datasheets, model cards, experiment logs, evaluation reports, monitoring reports, and retraining records.

5. Capability Maturity Model Integration (CMMI)

5.1 What is CMMI?

Capability Maturity Model Integration (CMMI) is a process-improvement model. It helps organizations improve how they plan, perform, measure, and improve work. CMMI does not directly certify that a particular software product is defect-free. Instead, it assesses the maturity of organizational processes that are expected to produce better and more predictable results.

5.2 Why CMMI is useful

- It makes project work less dependent on individual heroics and more dependent on repeatable processes.
- It encourages planning, measurement, risk management, quality assurance, and continuous improvement.
- It helps organizations estimate schedule, cost, and quality more reliably.
- It is especially useful for large organizations, government suppliers, defense projects, and regulated industries.

5.3 CMMI maturity levels

Level	Name	Simple meaning	Student example
1	Initial	Work is unpredictable and often reactive. Success depends heavily on individual effort.	A college project team codes at the last minute without plan, tests, or documentation.
2	Managed	Projects are planned, performed, measured, and controlled at the project level.	Team has sprint plan, task tracking, test cases, and version control.
3	Defined	Organization-wide standard processes are defined and used across projects.	All student project groups follow a common template for requirements, design, testing, and review.
4	Quantitatively Managed	Processes are measured and controlled using quantitative data.	Defect density, test coverage, lead time, and model performance drift are tracked statistically.
5	Optimizing	Organization continuously improves processes using data and innovation.	The team analyzes root causes of defects and improves pipeline automation and training data checks.

5.4 CMMI and software quality

CMMI improves quality indirectly. A mature process helps ensure that requirements are managed, risks are tracked, defects are analyzed, and improvements are institutionalized. However, a high maturity level should not be misunderstood as a guarantee of perfect software. Product testing, security assessment, user validation, and legal compliance are still necessary.

Quality problem	How process maturity helps
Frequent missed deadlines	Better estimation, planning, monitoring, and risk control.
Repeated defects	Root-cause analysis and defect-prevention process.
Poor communication	Standard documents, reviews, and defined responsibilities.
Uncontrolled changes	Change control and configuration management.
Weak evidence in disputes	Documented process records, test results, approvals, and traceability.

5.5 Limitations of CMMI

- CMMI can become paperwork-heavy if implemented mechanically.
- It evaluates process maturity, not automatically the actual quality or ethics of every product.
- Small teams may need a lightweight adaptation rather than a heavy process bureaucracy.
- Agile and DevOps teams can use maturity thinking, but they should avoid slowing down useful feedback loops.

5.6 Example: process maturity in model deployment

Low maturity practice	Mature practice
Dataset is collected without documentation.	Dataset source, license, consent, preprocessing, and limitations are documented.
Model version is overwritten repeatedly.	Model registry stores versions, metrics, parameters, and approval status.
Only accuracy is reported.	Accuracy, precision, recall, fairness, robustness, latency, and drift risk are measured.
Deployment happens manually.	CI/CD or MLOps pipeline tests the model and application before release.
No monitoring after deployment.	Performance, drift, errors, and security incidents are monitored.

6. Safety-Critical System

6.1 Meaning of safety-critical system

A safety-critical system is a system whose failure can cause death, serious injury, major environmental damage, or major loss of property. In such systems, software quality is not merely desirable; it is essential for protecting human life and public safety.

6.2 Examples of safety-critical systems

Domain	Example system	Possible harm if it fails
Healthcare	Infusion pump, radiotherapy machine, patient monitoring system.	Wrong medication dose, missed emergency alert, patient injury.
Transport	Aircraft control, railway signaling, autonomous driving assistance.	Collision, derailment, crash, injury or death.
Industrial control	Chemical plant control, power grid protection, nuclear plant monitoring.	Explosion, toxic release, blackout, environmental damage.
Defense and aerospace	Navigation, targeting, space mission control.	Mission loss, unintended damage, human harm.
Public infrastructure	Dam control, emergency alert system, traffic management.	Flooding, failure to warn citizens, traffic accidents.
AI-enabled systems	AI triage, autonomous vehicle perception, predictive maintenance for critical machinery.	Incorrect decisions affecting health, safety, or essential services.

6.3 Important safety terms

Term	Simple meaning
Hazard	A possible source of harm.
Risk	Combination of likelihood and severity of harm.
Safety requirement	A requirement written to prevent or control hazards.
Fail-safe	Design behavior that moves the system to a safe state when failure occurs.
Redundancy	Using backup components so one failure does not cause total system failure.
Fault tolerance	Ability to continue safe operation even when some faults occur.
Safety integrity	Confidence that a safety function will work when needed.
Assurance case	Structured argument with evidence showing that a system is acceptably safe.

6.4 Safety engineering activities

Activity	Simple explanation
Hazard identification	List what can go wrong and who/what can be harmed.
Risk analysis	Estimate severity and likelihood of each hazard.
Safety requirements	Define controls to prevent, detect, or reduce hazards.
Formal methods where needed	Use mathematical specification/verification for high-risk logic.
Independent verification and validation	Use independent reviewers/testers for high-assurance systems.
Fault injection and stress testing	Deliberately test failure conditions.
Configuration management	Control versions of code, hardware, models, data, and documents.
Incident learning	Study failures and near misses to improve the system.

6.5 Difference between safety and security

Concept	Main question	Example
Safety	Can the system harm people, property, or the environment if it fails?	A braking system must stop the vehicle safely.
Security	Can an attacker misuse, damage, or access the system?	An attacker must not remotely disable the braking system.
Safety-security connection	Security failures can become safety failures.	A hacked medical device can become unsafe even if its original design was safe.

6.6 AI/ML in safety-critical systems

AI/ML can support safety-critical functions, but it also creates challenges. ML models may be difficult to interpret, may fail on rare cases, and may change behavior when retrained. Therefore, high-risk AI systems need strong validation, monitoring, human oversight, and clear operational limits.

- Avoid using black-box AI as the only decision maker in high-risk situations unless the safety case is strong and domain rules allow it.
- Use human-in-the-loop or human-on-the-loop oversight where decisions affect life, health, or fundamental rights.
- Test models on edge cases, rare events, adversarial inputs, and changing environments.
- Define safe fallback behavior when the model is uncertain or when sensors fail.
- Maintain traceability of training data, model version, evaluation report, and deployment approval.

7. Quality Management Standards

7.1 Why quality management standards are needed

A standard provides an agreed way to define terms, processes, documents, measurements, and quality expectations. Standards help organizations communicate clearly, compare practices, train teams, perform audits, and show responsible development. Standards do not remove all risk, but they provide a disciplined framework for reducing and managing risk.

7.2 Important standards and frameworks

Standard/framework	Main purpose	Usefulness
ISO 9001:2015	Quality management system requirements for organizations.	Shows how organizations manage quality systematically and continually improve.
ISO/IEC/IEEE 90003:2018	Guidance for applying ISO 9001 to software acquisition, supply, development, operation, and maintenance.	Connects general quality management with software work.
ISO/IEC 25010:2023	Software and ICT product quality model.	Provides vocabulary for quality attributes such as reliability, security, maintainability, and usability.
ISO/IEC/IEEE 12207:2017	Common framework for software life-cycle processes.	Helps students understand development from conception to retirement.
IEEE 730	Software Quality Assurance process requirements.	Useful for planning and controlling SQA activities in projects.
ISO/IEC/IEEE 29119 series	Software testing concepts, processes, documentation, and techniques.	Provides structured testing language and process ideas.
CMMI	Process capability and maturity improvement model.	Explains how organizations become more predictable and continuously improve.
IEC 61508	Functional safety for electrical/electronic/programmable electronic safety-related systems.	Important for safety-critical software and embedded systems.

NIST SSDF SP 800-218	Secure software development practices to reduce vulnerability risk.	Helpful for secure SDLC and cyber law/ethics topics.
ISO/IEC 42001:2023	AI management system standard.	Relevant for responsible governance of AI-based systems.
ISO/IEC 23894:2023	Guidance on AI risk management.	Relevant for identifying and managing AI-specific risks.
ISO/IEC 25059:2023	Quality model for AI systems as an extension of SQuaRE.	Relevant for quality evaluation of AI systems beyond traditional software.
NIST AI RMF 1.0	Framework to manage AI risks to individuals, organizations, and society.	Useful for trustworthy AI, risk management, and governance.

7.3 ISO 9001 and ISO/IEC/IEEE 90003

ISO 9001 is a general quality management standard. It focuses on establishing, implementing, maintaining, and continually improving a quality management system. ISO/IEC/IEEE 90003 gives guidance on applying ISO 9001 to software-related activities such as acquisition, supply, development, operation, and maintenance.

Quality management idea	Meaning in software project
Customer focus	Understand user needs and legal/contractual expectations.
Leadership	Management must support quality, not only deadlines.
Process approach	Define and improve how work is performed.
Evidence-based decision making	Use metrics, test results, incidents, and audits rather than guesses.
Continual improvement	Learn from defects, feedback, and failures.

7.4 ISO/IEC 25010 and product quality model

ISO/IEC 25010 is useful because it gives a structured vocabulary for software quality. Instead of saying “the software is good,” a team can discuss specific qualities: functional suitability, reliability, performance efficiency, usability, security, compatibility, maintainability, and portability. The 2023 version addresses software and ICT product quality and is part of the SQuaRE family of standards.

7.5 IEEE 730 and software quality assurance plan

IEEE 730 is relevant to Software Quality Assurance (SQA). A good SQA plan usually defines quality responsibilities, review activities, audits, documentation, configuration management, testing coordination, defect tracking, supplier control, and reporting. In student projects, a simplified SQA plan can still be very useful.

7.6 Testing standards and quality evidence

Testing is a major source of quality evidence. ISO/IEC/IEEE 29119 provides general testing concepts and related standards for test process, documentation, techniques, and other areas. For students, the main lesson is that testing should be planned, traceable, repeatable, documented, and risk-based.

Testing type	Purpose
Unit testing	Tests small units such as functions, classes, or modules.
Integration testing	Checks whether modules work together correctly.
System testing	Tests the whole system against requirements.
Acceptance testing	Checks whether users or customers accept the system.
Regression testing	Checks that changes did not break earlier functions.
Security testing	Finds vulnerabilities and misuse paths.
Performance testing	Checks speed, scalability, and resource usage.
AI/ML evaluation	Checks model metrics, bias, robustness, calibration, and drift behavior.

7.7 AI management and risk standards

AI systems require governance because they can affect fairness, transparency, privacy, safety, and accountability. ISO/IEC 42001 gives requirements for an AI management system. ISO/IEC 23894 gives guidance for AI risk management. ISO/IEC 25059 provides a quality model for AI systems. NIST AI RMF is another major framework for managing AI risks to individuals, organizations, and society.

2. Impact of Information Technology

Information Technology means the use of computers, networks, software, databases, cloud platforms, mobile devices, and communication systems to collect, store, process, transmit, and use information. In modern society, IT is no longer only a support tool. It has become part of education, business, banking, healthcare, government, agriculture, transport, entertainment, and scientific research.

2.1 Positive impacts of IT

- **Faster communication:** Email, video calls, messaging platforms, and collaboration tools allow people to work together across distance.
- **Better access to information:** Students can use digital libraries, online courses, journals, tutorials, and government portals.
- **Automation of routine tasks:** Billing, payroll, ticket booking, attendance, inventory management, and report generation can be automated.
- **New services and business models:** E-commerce, digital payments, cloud computing, online education, telemedicine, and platform-based services have grown because of IT.
- **Improved public services:** Digital identity, online forms, e-governance portals, grievance systems, and direct benefit transfers can reduce paperwork and improve transparency.
- **Better decision-making:** Data analytics and AI can help organizations identify patterns, forecast demand, detect fraud, and support planning.
- **Inclusion opportunities:** Assistive technologies such as screen readers, speech-to-text, captioning, and translation tools can support persons with disabilities.

2.2 Negative or risky impacts of IT

- **Privacy loss:** Data about location, health, purchases, communication, and browsing can be misused if not properly protected.
- **Cybercrime:** Phishing, ransomware, identity theft, data breaches, and online fraud are major

risks in digital society.

- Job displacement and skill gaps: Automation can reduce demand for some routine jobs while increasing demand for digital skills.
- Misinformation: False or misleading content can spread quickly on social media and search platforms, especially in health and politics.
- Overdependence: Excessive reliance on digital systems can create problems when systems fail, networks go down, or data becomes unavailable.
- Digital stress: Constant notifications, online surveillance at work, and pressure to remain available can affect mental well-being.
- Algorithmic bias: AI systems trained on biased or incomplete data may produce unfair outcomes.

2.3 IT and quality of life

Quality of life means the overall well-being of people. IT can improve quality of life by saving time, improving access to services, helping people learn, enabling remote work, supporting healthcare, and connecting families. However, it can also create inequality, addiction-like use, surveillance, privacy risks, and social isolation if not used responsibly. OECD discussions on digital transformation emphasize that technology policy should support growth, well-being, inclusion, privacy, security, and trust.

Area of life	Possible improvement due to IT	Possible risk
Education	Online learning, digital libraries, language tools, AI tutoring.	Unequal internet/device access, plagiarism, distraction.
Work	Remote work, automation, data-driven decisions.	Job loss in routine tasks, monitoring stress, skill gap.
Health	EHRs, telemedicine, remote monitoring, health information.	Data breaches, misinformation, unequal access, overdiagnosis.
Government services	Online applications, transparency, faster service delivery.	Exclusion of digitally illiterate citizens, identity fraud.
Social life	Communication across distance, community support.	Cyberbullying, misinformation, privacy loss.

3. IT Investment and Productivity

IT investment means money and effort spent on hardware, software, networks, cloud services, cybersecurity, data platforms, training, and process redesign. Productivity generally means how much useful output is produced from a given amount of input such as time, labour, capital, and resources.

3.1 Why IT can increase productivity

- Speed: Computers can process transactions, records, and calculations faster than manual systems.
- Accuracy: Well-designed software reduces some human errors in repetitive tasks.
- Scalability: Cloud and automation allow services to serve many users at the same time.
- Coordination: Shared databases and workflow systems reduce duplication and delay.
- Analytics: Data mining, dashboards, and AI help managers make evidence-based decisions.
- Innovation: IT enables new products, online services, digital marketplaces, and personalized services.

3.2 The productivity paradox

Researchers have discussed a problem called the IT productivity paradox: organizations may spend heavily on computers and software, but productivity gains may not be immediately visible in official productivity statistics or business results. Brynjolfsson described this issue as a mismatch between visible computer investment and measured productivity improvements.

Buying technology is like buying a powerful machine. It does not automatically improve results unless people know how to use it, the work process is redesigned, and the organization measures the right outcomes.

For example, installing an EHR in a hospital may initially slow doctors because they must learn the system. Benefits appear only when workflows, training, interoperability, and usability improve.

3.3 Reasons IT investment may fail to improve productivity

- Poor planning: Technology is purchased without a clear problem statement.
- Lack of training: Users do not understand how to use the system effectively.
- Bad process design: Old paper-based processes are copied into software without improvement.
- Poor usability: Software is confusing, slow, or difficult to use.
- Data quality problems: Incomplete or wrong data leads to wrong decisions.
- Cybersecurity incidents: Breaches and downtime reduce trust and productivity.
- Wrong metrics: Benefits such as better quality, faster response, and improved customer satisfaction may not be captured by simple output measures.

Good IT investment practice	Why it matters
Start with a real problem	Prevents buying technology only because it is fashionable.
Redesign the process	Software should simplify and improve work, not reproduce inefficient manual steps.
Train users	Employees need skills and confidence to use new systems.
Measure outcomes	Track time saved, quality improved, errors reduced, and user satisfaction.
Protect security and privacy	A productive digital system must also be trustworthy.
Improve continuously	Feedback and updates are required after deployment.

AI investment must be evaluated with both technical and organizational metrics. Accuracy alone is not enough. For an AI chatbot in healthcare, evaluate safety, hallucination rate, privacy risk, bias, user trust, escalation to doctors, and actual reduction in workload.

4. Digital Divide

The digital divide refers to unequal access to digital technologies and unequal ability to use them. OECD describes digital divide in terms of unequal access to internet connectivity, devices, infrastructure, skills, and affordability, leading to unequal opportunities for information access and digital participation. The World Bank also links digital exclusion with wider development gaps.

4.1 Main types of digital divide

Type	Meaning	Example
Access divide	Some people do not have internet, devices, or electricity.	A rural student cannot attend online classes due to poor connectivity.
Affordability divide	Internet packs, devices, repairs, and software are too expensive.	A family shares one smartphone among several children.
Skills divide	People have access but do not know how to use digital tools safely.	A patient cannot use a health app or identify phishing messages.
Usage divide	Some users only consume entertainment and cannot use digital tools for education, jobs, or services.	A user can use social media but cannot fill online forms.
Quality divide	Speed, reliability, language support, and accessibility differ.	A website works only in English and excludes local-language users.
Data divide	Some groups are missing or underrepresented in datasets.	An AI system performs poorly for a region because training data is limited.

4.2 Causes of the digital divide

- Income inequality and affordability problems.
- Poor network infrastructure in rural or remote regions.
- Low digital literacy and lack of training.
- Language barriers and lack of local-language content.
- Gender, age, disability, and social barriers.
- Lack of trust in digital systems due to fraud, privacy risks, or poor service design.
- Weak accessibility design for persons with disabilities.

4.3 Effects of the digital divide

- Educational inequality: Students without devices or internet fall behind.
- Employment inequality: Digital skills are increasingly needed for jobs and entrepreneurship.
- Healthcare inequality: People without digital access may not benefit from telemedicine or online health records.
- Government service exclusion: Citizens may be unable to access online schemes or certificates.
- AI bias: If data from some communities is missing, AI systems may not work well for them.

Ways to reduce the digital divide

- Provide affordable broadband and public Wi-Fi in underserved areas.
- Support shared access points such as schools, libraries, community centres, and health centres.
- Teach digital literacy, cybersecurity hygiene, and online service use.
- Design websites and apps in local languages with accessibility features.
- Use inclusive datasets and fairness testing in AI/ML systems.
- Keep offline alternatives where necessary so citizens are not excluded.

5. Impact of IT on Healthcare Cost

IT can affect healthcare cost in two opposite ways. It can reduce some costs by improving coordination, reducing duplicate tests, enabling remote consultation, and automating records. It can also increase costs because hospitals must buy systems, train staff, maintain cybersecurity, migrate data, and manage compliance.

5.1 Ways IT can reduce healthcare costs

- Reduced travel cost for patients through telemedicine, especially for follow-up visits.
- Reduced duplication of tests when previous reports are available electronically.
- Faster claims, billing, registration, and appointment management.
- Better medicine management through e-prescriptions and alerts.
- Remote monitoring that may reduce unnecessary hospital visits for some chronic conditions.
- Better preventive care through reminders, screening alerts, and health education.

5.2 Ways IT can increase healthcare costs

- Initial cost of hardware, software, cloud infrastructure, and network upgrades.
- Training cost for doctors, nurses, technicians, and administrative staff.
- Maintenance cost for licenses, upgrades, cybersecurity, and technical support.
- Interoperability cost when old systems cannot communicate with new systems.
- Cybersecurity and privacy compliance costs.
- Risk of unnecessary tests if systems are poorly designed or create alert fatigue.

Cost area	Possible IT benefit	Possible risk
Registration and billing	Less paperwork and faster patient flow.	System downtime may stop service.
Diagnostics	Earlier access to reports and image sharing.	Duplicate digital systems may add cost.
Consultation	Remote follow-up can save travel and waiting time.	Unsuitable for emergencies or physical examination needs.
Drug management	E-prescriptions and interaction alerts.	Wrong data entry can lead to medication errors.
Public health	Digital surveillance and dashboards.	Privacy risk if data governance is weak.

6. Electronic Health Records (EHRs)

An Electronic Health Record is a digital record of a patient’s health information maintained by healthcare providers. It may include demographics, medical history, diagnoses, medicines, allergies, immunization, laboratory reports, radiology reports, progress notes, and treatment plans. ONC and CMS describe EHRs as patient-centred digital records that can make health information securely available to authorized users and support care-related activities [R10].

6.1 EHR vs EMR

Term	Meaning	Simple example
EMR: Electronic Medical Record	A digital version of records within one clinic or hospital.	A doctor's clinic keeps patient visit notes electronically.
EHR: Electronic Health Record	A broader digital record designed for sharing across authorized healthcare providers and systems.	A patient's lab report, prescription, and hospital discharge summary can be shared with consent.

Typical EHR workflow

Patient registration -> doctor consultation -> diagnosis and prescription -> lab/radiology orders -> reports uploaded -> follow-up care -> record shared with consent when needed.

In AI/ML settings, de-identified EHR data may also support research, prediction models, and public health analytics, but only with proper governance.

6.2 Benefits of EHRs

- Improved continuity of care because authorized providers can view previous records.
- Better decision support through alerts for allergies, drug interactions, or preventive care.
- Reduced paper storage and easier retrieval of old reports.
- Improved coordination among hospitals, labs, pharmacies, and insurance systems.
- Support for quality management, outcomes reporting, and public health planning.
- Patient portals can help patients access test results, prescriptions, and visit summaries.

6.3 Risks and challenges of EHRs

- Privacy breach: Health data is sensitive personal data and requires strong protection.
- Unauthorized access: Only permitted users should access records based on role and need.
- Interoperability problems: Systems may not exchange data correctly if standards are not followed.
- Data quality problems: Wrong or incomplete data can harm patient care and AI model accuracy.
- Usability problems: Poorly designed screens may increase workload and cause errors.
- Cybersecurity threats: Ransomware and data theft can disrupt hospitals and expose patients.

6.4 Indian context: ABDM and ABHA

India's Ayushman Bharat Digital Mission (ABDM) aims to build a digital health ecosystem. The official ABDM website states that citizens can generate an Ayushman Bharat Health Account (ABHA) and share health records through the digital ecosystem. A Press Information Bureau release dated 22 May 2026 reported that ABDM had crossed 100 crore health records linked with ABHA.

7. Use of Mobile and Wireless Technology in the Healthcare Industry

Mobile health, often called mHealth, refers to the use of mobile and wireless technologies for public health and healthcare. WHO materials describe mHealth as the use of mobile wireless technologies for public health, and digital health as a broader concept that includes mHealth, eHealth, and newer fields such as big data and AI [R7].

7.1 Examples of mobile and wireless healthcare

Technology	Use in healthcare	AI/ML possibility
Health apps	Appointment reminders, medicine reminders, wellness tracking.	Personalized reminders and risk prediction.
Wearables	Heart rate, sleep, steps, oxygen saturation, ECG in some devices.	Anomaly detection and early warning systems.
Remote patient monitoring	Patients send blood pressure, glucose, or oxygen readings from home.	Predictive analytics for chronic disease management.
Wireless sensors	ICU monitoring, fall detection, smart beds, temperature monitoring.	Real-time alerts and pattern recognition.
Mobile diagnostics	Portable ultrasound, smartphone-based imaging, field health data collection.	Computer vision support and triage assistance.
SMS/IVR health messaging	Vaccination reminders, maternal health messages, outbreak alerts.	Targeted communication based on risk groups.

7.2 Benefits

- Improves access for rural, elderly, disabled, or busy patients.
- Supports continuous monitoring instead of only hospital-based measurement.
- Helps doctors track chronic diseases such as diabetes and hypertension.
- Saves travel time and may reduce unnecessary hospital visits.
- Enables public health communication at scale.
- Can generate data for AI-based disease prediction and personalized care.

7.3 Risks and limitations

- Device accuracy varies; consumer devices are not always clinical-grade medical devices.
- Apps may collect excessive personal data or share it with third parties.
- Wireless devices can be vulnerable to hacking or spoofing if not secured.
- Poor connectivity limits use in remote areas.
- False alarms can cause anxiety, and missed alarms can be dangerous.
- AI models built on wearable data may be biased if training data excludes certain age, gender, or health groups.

Cybersecurity practices for mobile health systems

- Use encryption for data in transit and at rest.
- Apply strong authentication and role-based access.
- Collect only necessary data and explain consent clearly.
- Regularly update apps, firmware, and servers.
- Audit access logs and detect abnormal access patterns.
- Perform safety testing before clinical use.

8. Telemedicine

Telemedicine means delivering clinical healthcare services when the patient and healthcare professional are not in the same physical place. WHO describes telemedicine as delivery of healthcare services over distance, with potential to improve clinical management and extend service coverage.

8.1 Telemedicine vs telehealth

Term	Meaning
Telemedicine	Remote clinical care such as video consultation, audio consultation, text consultation, remote diagnosis, and follow-up.
Telehealth	A broader term that can include clinical care, health education, public health services, remote monitoring, and administrative healthcare services.

8.2 Common modes of telemedicine

Mode	Explanation	Example
Video consultation	Doctor and patient interact through video call.	Dermatology follow-up or mental health counselling.
Audio consultation	Consultation through phone or voice call.	Follow-up for stable chronic conditions.
Text consultation	Messages, chat, or email-based communication.	Sharing reports and getting guidance where appropriate.
Store-and-forward	Medical data is captured and sent for later review.	X-ray, scan, or skin image sent to a specialist.
Remote monitoring	Device data is sent from patient location to healthcare provider.	Blood pressure monitoring for a cardiac patient.

8.3 When telemedicine is useful

- Follow-up consultations where physical examination is not essential.
- Chronic disease management and medicine review.
- Mental health counselling and health education.
- Specialist consultation for remote or underserved areas.
- Triage to decide whether in-person care or emergency care is needed.
- Public health emergencies where physical travel is difficult.

8.4 When telemedicine may not be suitable

- Emergency situations requiring immediate physical intervention.
- Cases requiring physical examination, surgery, or procedures.
- Poor internet/audio/video quality that prevents proper assessment.
- Patients unable to describe symptoms or provide reliable information.
- Situations where identity, consent, or medical record quality is uncertain.

8.5 Telemedicine in India

The Government of India issued Telemedicine Practice Guidelines on 25 March 2020. A PIB release states that these guidelines provide a framework for the practice of telemedicine. The guidelines cover aspects such as mode of communication, patient consent, identification of the registered medical practitioner and patient, documentation, privacy, and responsibilities of the practitioner.

9. Medical Information Websites

Medical information websites provide health-related information about diseases, symptoms, medicines, prevention, wellness, treatment options, and healthcare services. They can help patients learn, but they can also spread misinformation if content is not reliable.

9.1 Benefits of medical information websites

- Patients can learn about symptoms, prevention, and healthy behaviour.
- Students and citizens can access public health information quickly.
- Doctors can share patient education resources after consultation.
- Government portals can provide vaccination, disease prevention, and health scheme information.
- Reliable websites can reduce fear by explaining medical terms in simple language.

9.2 Risks of medical information websites

- Misinformation or outdated content can lead to harmful self-treatment.
- Commercial bias may promote products or tests unnecessarily.
- Symptom checkers may create anxiety or false confidence.
- Users may ignore professional medical advice after reading online content.
- Health data entered into websites or apps may be collected or shared without clear consent.

9.3 How to identify reliable medical information online

Question to ask	Why it matters
Who runs the website?	Government, university, hospital, or recognized medical organization websites are usually more trustworthy.
Who wrote or reviewed the content?	Medical content should be written or reviewed by qualified professionals.
Is the date updated?	Medical knowledge changes; old content may be unsafe.
Are sources cited?	Reliable sites explain where the information comes from.
Is there advertising or product promotion?	Commercial pressure may bias information.
Does it ask for personal health data?	Users should understand privacy policy and consent before sharing data.
Does it replace a doctor?	Reliable sites usually advise consulting qualified healthcare professionals.

10. Ethical, Legal, and Cybersecurity Issues

Because this topic belongs to Cyber Law and Ethics, students must connect technology benefits with ethical responsibility. IT can improve productivity and health, but it can also harm people if data, algorithms, and systems are misused.

Issue	Meaning	Example in healthcare/IT
Privacy	People should have control over sensitive personal data.	A hospital should not reveal patient records to unauthorized users.
Consent	Users should understand what data is collected and how it is used.	A health app asks permission before sharing records.
Confidentiality	Professionals must keep patient or user information secret unless lawfully permitted.	A doctor cannot casually share patient screenshots.
Security	Systems must protect data from unauthorized access or attack.	EHR databases need encryption, access control, and backups.
Accountability	Someone must be responsible when a digital system causes harm.	Who is responsible if an AI triage tool gives unsafe advice?
Transparency	Users should know how systems make important decisions where possible.	A hospital explains how a risk score is used.
Fairness	Systems should not discriminate unfairly.	An AI model should be tested across gender, age, language, and region.
Accessibility	Digital services should be usable by persons with disabilities and low digital literacy.	A portal supports screen readers and local languages.

10.1 Cybersecurity controls for digital health and public IT systems

- Authentication: Confirm that users are who they claim to be.
- Authorization: Give users only the access required for their role.
- Encryption: Protect data while stored and while transmitted.
- Audit logs: Record who accessed what data and when.
- Backups and disaster recovery: Keep services running after failures or cyberattacks.
- Patch management: Update software to fix known vulnerabilities.
- User training: Teach staff to detect phishing and handle data responsibly.
- Incident response: Prepare a plan for breaches, ransomware, downtime, or data loss.

2. Social Networking Websites

2.1 Meaning

A social networking website is an online platform that allows users to create profiles, connect with other users, communicate, share content, join groups or communities, and interact through likes, comments, messages, stories, live streams, or similar features. Examples include general social networks, professional networking sites, video-sharing platforms, microblogging services, image-sharing platforms, online communities, gaming networks, and messaging-based social platforms.

The core idea is not only content sharing but relationship building. The platform maintains a social graph: a structure showing users, their connections, groups, pages, interests, and interactions. In AI/ML terms, this graph can be used for recommendation, fraud detection, influence analysis, community detection, and safety monitoring.

2.2 Main Features of Social Networking Websites

Feature	Simple explanation	AI/ML relevance
User profile	A digital identity containing name, photo, bio, education, interests, posts, and links.	Identity matching, profile recommendation, impersonation detection, privacy-risk scoring.
Friend/follower network	Users connect with friends, followers, groups, channels, or pages.	Graph neural networks, community detection, and link prediction.
News feed or timeline	A personalised stream of posts, images, videos, advertisements, and updates.	Recommendation systems, ranking algorithms, engagement prediction.
Content sharing	Users upload text, images, videos, audio, files, live streams, and links.	NLP, computer vision, audio analysis, harmful-content moderation.
Messaging	Private or group communication through text, voice, video, or media sharing.	Spam detection, abuse detection, encryption and privacy design.
Groups and communities	Spaces for people with similar interests to discuss, learn, organise, or collaborate.	Topic modelling, sentiment analysis, community moderation.
Privacy settings	Controls that decide who can see, comment, message, tag, or share user content.	Privacy engineering, default-setting design, risk prediction.
Advertising and analytics	Platforms use data to show ads and measure engagement.	Targeting models, conversion prediction, fairness and transparency concerns.

2.3 Types of Social Networking Platforms

- Personal social networks: used for friendship, family updates, photos, short videos, and public posting.
- Professional networks: used for career identity, recruitment, professional learning, networking, and industry discussions.
- Media-sharing platforms: focused on videos, images, live streams, podcasts, reels, and creative work.
- Discussion communities: forums and community platforms where people discuss topics, solve problems, or share knowledge.
- Messaging-based social platforms: private or group communication, often used for communities, customer support, and informal learning.
- Gaming and virtual-world platforms: social interaction happens through avatars, voice chat, digital goods, virtual events, and multiplayer experiences.

2.4 Benefits of Social Networking

- Communication: people can connect quickly across distance.
- Learning: students can access tutorials, discussion groups, open lectures, research updates, and peer communities.
- Employment: users can discover internships, jobs, hackathons, scholarships, and professional mentors.
- Civic participation: people can spread awareness about social causes, disaster relief, public health, and government services.
- Business growth: companies can communicate directly with customers and study public feedback.
- Creativity: users can publish art, writing, music, code, educational videos, and innovations.

2.5 Risks of Social Networking

- Privacy loss: personal details, location, photos, contacts, and habits can be exposed.
- Identity misuse: fake profiles can impersonate people or organisations.
- Harassment: bullying, stalking, threats, doxxing, and hate speech can occur.
- Misinformation: false or manipulated content can spread quickly.
- Addictive design: infinite scroll, notifications, and personalised feeds may encourage excessive use.
- Data profiling: platforms may infer sensitive interests or behaviours from activity patterns.
- Legal risk: users may face consequences for unlawful posts, defamatory claims, copyright misuse, or non-consensual sharing.

3. Business Applications of Online Social Networking

Businesses use social networking not only for advertising but also for communication, collaboration, analytics, market research, recruitment, brand building, and customer support. Government business guidance notes that social media can help businesses promote products, connect with customers, build trust, advertise, sell, drive website traffic, and understand competitors.

Business application	How it works	Example for AI/ML students
Digital marketing and brand awareness	A company posts content, videos, product updates, educational posts, and campaigns to reach potential customers.	Use NLP to analyse comments and find whether a campaign is receiving positive, negative, or neutral sentiment.
Customer service	Customers ask questions or complain through comments, chatbots, or direct messages.	Build a chatbot that classifies complaints and routes them to the right department.
Social listening	The company monitors public mentions, reviews, hashtags, and discussion trends.	Use topic modelling to identify common product problems mentioned by customers.
Recruitment and employer branding	Companies share job openings, employee stories, internship posts, and technical culture.	Use fair screening tools carefully; avoid biased automated selection based on gender, caste, location, or college name.
Influencer and creator collaboration	Brands collaborate with creators to reach target communities.	Detect fake engagement, bot followers, and artificial comment patterns.
Social commerce	Users discover and buy products directly through posts, videos, live streams, or platform shops.	Build recommendation models while respecting privacy and transparency.
Internal collaboration	Organisations use social intranets or enterprise communication tools for teams and knowledge sharing.	Design search and summarisation systems for internal knowledge bases with access control.
Crisis communication	During product failures, data breaches, or public complaints, organisations issue updates and respond publicly.	Use anomaly detection to identify sudden spikes in complaints or misinformation.

3.1 Advantages for Business

- Lower entry cost compared with traditional advertising, especially for small businesses.
- Fast feedback from customers and communities.
- Direct relationship building with users.
- Better understanding of market trends and customer needs.
- Global reach and niche targeting.
- Measurable engagement through views, clicks, shares, conversions, and sentiment.

3.2 Risks for Business

- Reputation damage from insensitive posts, poor moderation, data breaches, or false claims.
- Legal liability for misleading advertising, privacy violations, copyright misuse, or failure to handle harmful content properly.
- Security threats such as account takeover, phishing through official-looking pages, and fake customer-support handles.
- Dependence on platform algorithms that may change without warning.
- Biased analytics if data is incomplete, manipulated by bots, or not representative of real customers.

4. Social Networking Ethical Issues

Ethics asks: even if something is technically possible or legally permitted, is it fair, respectful, safe, and socially responsible? Social networking raises ethical issues because it joins personal identity, public speech, private data, AI-based ranking, business incentives, and social influence in one environment.

Ethical issue	What it means	Example
Privacy and consent	Users may not understand how much data is collected, inferred, shared, or sold.	An app asks for contact access even when contact access is not necessary for its main function.
Digital footprint	Posts, screenshots, metadata, and archived content may remain long after deletion.	A careless post during college affects job screening years later.
Misinformation and manipulation	False, misleading, or edited content can influence opinions or cause harm.	A deepfake video spreads before it is verified.
Algorithmic bias	Recommendation or moderation systems may treat groups unfairly or amplify harmful content.	A content moderation model fails for Indian languages or dialects because training data is limited.
Addiction and attention design	Platform features may be designed to maximise time spent rather than user well-being.	Endless scrolling and notifications make it difficult for students to focus.
Harassment and abuse	Users may face threats, insults, bullying, stalking, doxxing, or sexual harassment.	A student is targeted in a group chat after disagreeing with others.
Intellectual Property	Users may upload copyrighted music, images, code, or notes without permission.	A page reposts another student's project demo without credit.
Transparency	Users may not know whether content is produced by a human, bot, paid influencer, or AI system.	A product review is secretly sponsored or generated by bots.

5. Cyberbullying

UNICEF defines cyberbullying as bullying through digital technologies. It can occur on social media, messaging platforms, gaming platforms, and mobile phones, and it is usually repeated behaviour aimed at scaring, angering, or shaming the targeted person.

5.1 Common Forms of Cyberbullying

- sending repeated insulting, threatening, or humiliating messages;
- posting rumours, edited photos, or embarrassing content about someone;
- creating fake accounts to mock or impersonate a person;
- excluding someone deliberately from online groups to humiliate them;
- sharing private chats, photos, or personal information without consent;
- encouraging others to attack, troll, or mass-report a person;
- Using memes or jokes to repeatedly target a person's body, language, gender, caste, religion, disability, academic performance, or personal life.

5.2 Why Cyberbullying is Serious

Cyberbullying can spread quickly, remain searchable, and reach the victim even at home. It may cause fear, anxiety, and loss of confidence, academic decline, social isolation, and reputational harm. The online nature of the harm does not make it less serious than face-to-face bullying.

5.3 What a Student Should Do

- Do not retaliate with abusive messages; it can worsen the situation.
- Save evidence: screenshots, profile links, URLs, dates, times, and usernames. Do not edit evidence.
- Block or restrict the abuser using platform tools where appropriate.
- Report the content to the platform or group administrator.
- Tell a trusted teacher, parent, mentor, college authority, or counsellor.
- For serious cybercrime, report through the National Cyber Crime Reporting Portal. The portal is a Government of India facility for reporting cybercrime complaints online, with special focus on crimes against women and children.

6. Cyberstalking

Cyberstalking means repeated, unwanted online attention, monitoring, contact, tracking, intimidation, or surveillance that makes a person feel unsafe. It may include direct messages, fake accounts, repeated friend requests, location tracking, spyware, threats, monitoring online activity, or contacting friends and family to pressure the victim.

6.1 Cyberstalking vs Cyberbullying

Point	Cyberbullying	Cyberstalking
Main purpose	Humiliation, social attack, group targeting, public shaming.	Monitoring, control, intimidation, repeated unwanted contact.
Typical mode	Posts, comments, group chats, memes, rumours.	Messages, tracking, fake accounts, surveillance, threats.
Visibility	Often public or semi-public.	Often private, hidden, or repeated across accounts.
Victim experience	Embarrassment, fear, social isolation.	Fear, loss of safety, constant monitoring, anxiety.
Overlap	A cyberbully may also become a cyberstalker if the behaviour becomes repeated unwanted pursuit or monitoring.	A cyberstalker may use bullying techniques such as public shaming or threats.

6.2 Legal Context in India

Under the Bharatiya Nyaya Sanhita, 2023, stalking includes repeated unwanted contact and monitoring the use of the internet, email, or other electronic communication by a woman in the circumstances described in the section. Other online acts connected with cyberstalking may also involve provisions of the Information Technology Act, 2000, such as identity theft, cheating by personation, violation of privacy, and transmission of unlawful explicit material depending on the facts.

6.3 Technical Methods Used by Cyberstalkers

- creating multiple fake accounts after being blocked;
- tracking location through shared posts, check-ins, metadata, or leaked live location;
- installing spyware or misusing device access;
- using leaked passwords to access accounts;
- impersonating the victim to contact others;
- scraping photos, posts, or contact lists for intimidation or blackmail;
- Using social engineering to obtain personal information from friends or relatives.

6.4 Prevention Tips

- Use strong, unique passwords and multi-factor authentication.
- Review privacy settings and limit who can see phone numbers, email addresses, friend lists, and location.
- Avoid sharing real-time location or daily routine publicly.
- Check active login sessions and revoke unknown devices.
- Do not install unknown APKs or links received through messages.
- Keep screenshots and URLs if harassment occurs; evidence helps reporting and investigation.

7. Online Sexual Predators and Grooming Risks

In academic and safety contexts, the term “sexual predators” generally refers to offenders who use online platforms to exploit, manipulate, groom, threaten, or sexually abuse others, especially children and vulnerable users. UNICEF notes that online sexual exploitation and abuse is a serious threat to children in digital environments.

7.1 What is Online Grooming?

Online grooming is a process in which an offender builds trust with a child or vulnerable person to manipulate them into sexual exploitation, secrecy, blackmail, or unsafe meetings. It may begin with friendly conversation, gifts, compliments, gaming help, fake emotional support, or pretending to be of the same age.

7.2 Warning Signs

- A stranger quickly asks for private chat, video call, personal photos, or secrecy.
- The person gives gifts, game credits, money, or emotional attention to gain trust.
- The person pressures the user not to tell parents, teachers, friends, or moderators.
- The person asks for location, school, routine, private images, or personal problems.
- The person becomes angry, threatening, or blackmailing when the user refuses.
- The person uses fake age, fake identity, fake profile photos, or multiple accounts.

7.3 Legal and Reporting Context

The Information Technology Act, 2000 includes Section 67B, which deals with publishing or transmitting material depicting children in sexually explicit acts in electronic form and related conduct [R9]. The National Cyber Crime Reporting Portal provides reporting categories focused on crimes against women and children, including child sexual exploitative and abuse material [R7].

7.4 Role of AI/ML in Protection

- detecting grooming language patterns in chats, while protecting privacy and avoiding false accusations;
- finding networks of fake accounts and repeat offenders;
- detecting known illegal material using hash-matching technologies used by authorised platforms;
- identifying suspicious behaviour patterns such as rapid movement of conversations from public to private channels;
- Supporting moderators with risk scores, while final decisions should involve human review in sensitive cases.

8. Uploading Inappropriate Material

Uploading inappropriate material means posting, sharing, and forwarding, storing, or distributing content that is harmful, unlawful, offensive, privacy-violating, or unsuitable for the audience or platform. The same content may be inappropriate for ethical reasons, legal reasons, institutional-policy reasons, or safety reasons.

8.1 Types of Inappropriate Material

Type	Why it is harmful	Examples
Non-consensual private content	Violates privacy, dignity, consent, and safety.	Private photos, intimate images, screenshots of private chats, medical or family details.
Obscene or sexually explicit material	May violate law and platform rules, especially if non-consensual or involving children.	Explicit videos, manipulated images, forwarding illegal content.
Hate speech and targeted abuse	Can intimidate or degrade people based on identity.	Abuse based on caste, religion, language, gender, disability, region, or community.
Defamation and false allegations	Can damage reputation and career.	False claims about a person, teacher, student, company, or public figure.
Violence and self-harm content	Can traumatize users or encourage harm.	Graphic violence, dangerous challenges, instructions encouraging harm.
Personal data and doxing	Exposes people to stalking, fraud, and physical risk.	Phone numbers, addresses, IDs, bank details, location, live routine.
Malware or phishing links	Can steal data, money, accounts, or device access.	Fake login links, unknown APK files, fake customer-support links.
Copyrighted or confidential material	Violates IP rights, academic integrity, or organisational secrecy.	Paid books, exam papers, company source code, leaked project documents.

8.2 Legal Context

Indian law contains several provisions that may become relevant depending on the content. The IT Act includes provisions related to identity theft, cheating by personation, violation of privacy, obscene material, sexually explicit material, child sexual abuse-related electronic material, and breach of confidentiality or privacy. The Digital Personal Data Protection Act, 2023 recognises protection of digital personal data and defines duties and rights connected with personal data processing.

The IT Intermediary Guidelines and Digital Media Ethics Code Rules, 2021 set due-diligence and grievance-related obligations for intermediaries, including social media intermediaries. For students, the practical lesson is that platform responsibility does not remove user responsibility: a user may still be responsible for what they upload or forward.

8.3 Before Uploading: The THINK Test

- T - True: Is the content verified and not misleading?
- H - Helpful: Does it help anyone, or is it only meant to mock or harm?
- I - Informed consent: Do all identifiable people agree to this posting?
- N - Necessary: Is it necessary to share publicly, or can it be kept private?
- K - Kind and lawful: Is it respectful, safe, and within law and platform rules?

9. Online Virtual Worlds

An online virtual world is a computer-generated, networked environment where users interact through avatars, objects, places, chat, voice, gestures, virtual goods, games, classrooms, simulations, or events. Some virtual worlds are 2D or game-like; others use 3D graphics, virtual reality (VR), augmented reality (AR), mixed reality (MR), blockchain assets, or metaverse-like experiences.

9.1 Key Features

- Persistence: the virtual space may continue to exist even after one user logs out.
- Avatar identity: users act through digital bodies or characters.
- Real-time interaction: users communicate through voice, text, gestures, or shared activities.
- Virtual economy: platforms may include virtual currency, skins, NFTs, digital land, goods, or services.
- User-generated content: users may build objects, rooms, games, assets, or events.
- Immersion: VR/AR devices can make the environment feel spatial and embodied.
- Platform governance: rules, moderation, access control, reporting tools, and community norms shape behavior.

Examples of Virtual-World Activities

- multiplayer games and social gaming spaces;
- virtual classrooms and labs;
- metaverse-style events, exhibitions, concerts, and conferences;
- digital-twin simulations of factories, hospitals, campuses, or cities;
- virtual offices and remote collaboration rooms;
- training simulations for aviation, medicine, defence, engineering, or disaster management;
- 3D design, architecture walkthroughs, and product demonstrations.

9.2 Why Virtual Worlds Raise New Ethical Questions

Virtual worlds combine speech, behaviour, avatars, data collection, biometric signals, spatial tracking, digital assets, and real-time interaction. As a result, traditional online harms can feel more intense because users experience them in immersive, embodied environments. UNESCO highlights that online and virtual worlds require attention to privacy, security, ethical obligations, and risks in digital environments.

10. Crime in Virtual Worlds

A crime in a virtual world may involve real-world harm, illegal access, fraud, harassment, theft of digital assets, grooming, exploitation, identity misuse, or cyber attacks. INTERPOL has warned that metaverse-like platforms may enable both existing crimes in new forms and emerging “metacrimes,” including grooming, radicalisation, cyber-physical threats, theft of virtual property, trespass in private virtual spaces, and robbery from an avatar.

Virtual-world risk	How it may happen	Possible controls
Account takeover	Attackers steal passwords, session tokens, or MFA codes to control avatars or assets.	MFA, login alerts, device management, phishing-resistant authentication.
Virtual asset theft	Digital goods, skins, tokens, or virtual currency are stolen or transferred through fraud.	Transaction verification, anomaly detection, fraud monitoring, user warnings.
Harassment and abuse	Users threaten, follow, sexually harass, or verbally abuse others through avatars, chat, or voice.	Reporting tools, block/mute, safe zones, human moderation, and behavior logs.
Impersonation and deep fake avatars	Attackers create avatars or voices resembling real people to deceive users.	Verified identities for sensitive contexts, watermarking, liveness checks, and user education.
Grooming and child exploitation	Offenders contact minors in games or virtual worlds and move them to private channels.	Age-appropriate design, parental controls, grooming detection, trained moderators, reporting.
Fraud and scams	Fake investments, fake digital goods, fake jobs, romance scams, or phishing in virtual spaces.	Scam detection, platform warnings, trusted escrow, and user verification.
Doxing and privacy invasion	Users reveal or collect real names, addresses, location, biometrics, voice, or behavioural data.	Privacy defaults, minimised data collection, consent, data protection controls.
Malware and malicious links	Attackers share fake game mods, NFT links, event invites, or APK files.	Link scanning, sandboxing, user education, safe download policies.

10.1 Evidence Issues in Virtual-World Crimes

- Evidence may include screenshots, screen recordings, chat logs, transaction IDs, account IDs, timestamps, IP logs, device logs, and platform moderation records.
- Avatars may not reveal real-world identity; investigation may need platform cooperation.
- Jurisdiction can be complex because users, servers, companies, and victims may be in different countries.
- AI-generated avatars, bots, and deepfakes can make attribution difficult.
- Students should not try to hack, retaliate, or collect illegal material. Preserve safe evidence and report through proper channels.

11.1 Educational Uses

Educational use	How it helps	Example
Virtual laboratories	Students can conduct simulated experiments safely and repeatedly.	Physics, chemistry, robotics, cybersecurity labs.
Medical and emergency training	Learners practise procedures and decisions without risking patients.	Anatomy visualisation, disaster response, first-aid simulations.
Historical and cultural tours	Students explore reconstructed places and events.	Ancient cities, museums, archaeological sites.
Engineering design	Students inspect 3D models and test designs collaboratively.	Bridge design, factory layout, and product prototype.
Language and communication practice	Learners interact with avatars in realistic conversation scenarios.	Interview practice, group discussion, and cross-cultural communication.
Inclusive learning	Students who cannot travel may join remote immersive activities.	Virtual campus tour, remote lab demonstration.

11.2 Limitations in Education

- High cost of devices, network bandwidth, content creation, and maintenance.
- Accessibility barriers for students with disabilities, motion sickness, or low-end devices.
- Privacy risks from voice, video, location, movement, gaze, and biometric-like data.
- Need for teacher training and strong lesson design; technology alone does not guarantee learning.
- Possibility of distraction, harassment, or unsafe interactions if moderation is weak.

11.3 Business Uses

Business use	Description	Example
Virtual meetings and offices	Teams meet in 3D rooms or collaborative spaces.	Remote design review with 3D models.
Training and simulation	Employees practise complex or risky tasks safely.	Safety training for factory workers or pilots.
Virtual showrooms	Customers inspect products in 3D before buying.	Car showroom, furniture preview, real-estate walkthrough.
Digital twins	A virtual model mirrors a real system for monitoring or testing.	Smart factory, power plant, logistics warehouse.
Marketing and events	Brands host product launches, exhibitions, or interactive experiences.	Virtual hackathon, career fair, campus placement event.
Customer support	Support staff guide users through product setup in virtual spaces.	Repair training or guided troubleshooting.

11.4 Ethical Business Use of Virtual Worlds

- Collect only necessary data and clearly explain what is collected.
- Do not use manipulative design to pressure users into purchases.
- Provide safety controls such as block, mute, personal boundaries, and reporting.
- Protect minors and vulnerable users with age-appropriate design.
- Use accessible design for people with disabilities and low-bandwidth users.
- Audit AI systems for bias, false positives, and privacy risks.

12. Indian Legal and Regulatory Framework: Quick Map

The following table is a simplified educational map. Real cases depend on facts, evidence, jurisdiction, and current law. Students should use official sources and qualified legal advice for actual matters.

Issue	Relevant source / legal idea	Simple meaning
Cybercrime reporting	National Cyber Crime Reporting Portal	Government of India online portal for reporting cybercrime complaints, with special focus on crimes against women and children.
Identity theft and impersonation	IT Act, 2000 provisions on identity theft and cheating by personation	Misusing another person's identity or pretending to be someone else online can have legal consequences.
Violation of privacy	IT Act Section 66E	Capturing, publishing, or transmitting certain private images without consent may be punishable.
Obscene/sexually explicit material	IT Act Sections 67 and 67A	Publishing or transmitting unlawful obscene or sexually explicit content in electronic form may be punishable.
Child sexual abuse-related material	IT Act Section 67B	Electronic material depicting children in sexually explicit acts and related conduct is treated seriously.
Cyberstalking	Bharatiya Nyaya Sanhita, 2023, Section 78	The law recognises stalking that includes monitoring internet, email, or electronic communication in the circumstances described in the section.
Data privacy	Digital Personal Data Protection Act, 2023	Personal data should be processed lawfully, with rights and obligations for individuals and organisations.
Intermediary responsibility	IT Intermediary Guidelines and Digital Media Ethics Code Rules, 2021	Intermediaries, including social media intermediaries, must follow due diligence and grievance-related obligations.

2. Introduction to Ethics of IT Organizations

An IT organization may be a software company, cloud-service provider, data-centre operator, social-media platform, AI start-up, cybersecurity firm, government IT department, or IT division inside a bank, hospital, university, or manufacturing company. Such organizations handle data, software, networks, digital services, algorithms, and sometimes critical infrastructure. Therefore, their decisions can affect privacy, employment, public safety, fairness, the environment, and social trust.

Ethics of an IT organization means the principles and practices that guide the organization to act responsibly, lawfully, fairly, and transparently while developing and using information technology. It includes what the organization should do even when the law is silent or when a decision is technically possible but socially harmful.

Simple example

- A company can technically monitor every keystroke of a remote software developer. However, doing so without transparency may violate trust and privacy. Ethical practice requires a clear policy, proportional monitoring, employee awareness, and respect for personal data.
- An AI company may be able to train a model on user data, but ethical practice requires lawful processing, consent or another valid basis, data minimization, security, and fairness checks.

3. Key Ethical Issues in IT Organizations

The following issues appear repeatedly in IT organizations.

Issue	Meaning	AI/ML or software example
Privacy and data protection	Organizations collect personal data such as names, addresses, health information, location, payment data, logs, and user behaviour. Ethical duties include collecting only necessary data, using it for stated purposes, protecting it, and deleting it when no longer needed.	A recommendation system should not secretly use sensitive user data unless lawfully permitted and ethically justified.
Cybersecurity and safety	Weak security can expose users, employees, and clients to fraud, identity theft, ransomware, or service disruption. Ethical organizations invest in secure design, vulnerability management, access control, and incident response.	A cloud API must use strong authentication, logging, encryption, and secure coding reviews.
Accuracy and reliability	Software errors can cause financial loss, medical harm, or public inconvenience. AI models may make wrong predictions. Ethical practice requires testing, validation, monitoring, and clear communication of limitations.	A medical AI classifier should not be released without clinical validation and human oversight.
Fairness and non-discrimination	Systems can discriminate if data, design, or policies are biased. Ethical IT organizations test systems for unfair outcomes and provide ways to challenge harmful decisions.	A hiring algorithm should be checked for gender, caste, disability, or language bias.
Transparency and accountability	Users and affected people should know who is responsible for decisions, how complaints can be made, and how automated decisions are governed.	An AI chatbot used in customer support should disclose that users are interacting with an automated system when relevant.
Intellectual property and licensing	Organizations must respect copyrights, software licences, open-source obligations, patents, trade secrets, and confidential information.	A developer should not copy proprietary source code from a former employer into a new project.

Conflicts of interest	Employees should not use their position for improper personal gain or favour a vendor in return for gifts, payments, or future employment.	A procurement engineer should disclose a personal relationship with a vendor.
Employee monitoring and workplace dignity	Security monitoring is sometimes necessary, but excessive surveillance can harm trust and autonomy. Policies should be clear, proportionate, and lawful.	Remote-work monitoring should focus on security and outcomes, not intrusive personal surveillance.
Environmental responsibility	Data centres, devices, AI training, and e-waste have environmental impact. Ethical organizations reduce energy use, improve device life, recycle responsibly, and design efficient systems.	An AI team can use smaller models, pruning, quantization, and efficient inference to reduce energy use.

4. Nontraditional Workers

A traditional worker is usually a full-time employee working directly for one employer, often with a fixed workplace and stable employment relationship. In contrast, nontraditional workers use different work arrangements. The International Labour Organization uses the broader idea of non-standard employment for work arrangements that differ from standard employment, such as temporary work, part-time or on-call work, temporary agency work, multiparty employment, disguised employment, and dependent self-employment.

Type	Meaning	Example
Freelancers / independent contractors	Work for clients on specific projects. They may choose their working hours and tools, but may have less job security.	A freelance UI designer builds screens for a mobile app.
Remote workers	Work away from the employer office using digital tools. They may be full-time employees or contractors.	A machine-learning engineer works from home and connects to cloud notebooks.
Gig/platform workers	Get tasks through digital platforms, often paid per task or project.	A data-labeling worker tags images for a computer vision dataset.
Part-time or on-call workers	Work fewer hours or only when needed.	A part-time cybersecurity analyst reviews weekly vulnerability reports.
Consultants	Provide expert advice for a limited period.	A cloud architect advises on migration to AWS/Azure/GCP.
Temporary agency workers	Are employed by an agency but work for a client organization.	A testing engineer is assigned to a software company through a staffing agency.

4.1 Ethical Issues Related to Nontraditional Workers

- Fair treatment: Nontraditional workers should not be treated as disposable human resources. They deserve respect, safe working conditions, timely payment, and clear expectations.
- Access and security: Temporary or remote access to systems must follow least-privilege principles. Access should be removed when the work ends.
- Confidentiality: Contractors may work with several clients. They must protect each client's confidential data and avoid conflicts of interest.
- Skill development: Organizations should not exclude nontraditional workers from essential security and ethics training if they handle sensitive systems.
- Algorithmic management: Platforms may assign work, rate workers, or deactivate accounts using algorithms. Ethical design requires transparency, appeal mechanisms, and protection against unfair automated decisions.
- Data-labeling ethics: AI/ML projects often depend on low-paid data labelers. Ethical organizations should provide fair pay, clear instructions, psychological safeguards for harmful content, and quality feedback.

Example: data annotation workforce

- A company building a hate-speech classifier may ask contract workers to label offensive content. Ethical management requires warning workers about distressing material, limiting exposure, providing support, paying fairly, and not hiding the nature of the work.
- Quality problems in labeling are not only technical; they may arise from unclear guidelines, worker fatigue, language mismatch, or unfair payment models.

5. Contingent Workers

Contingent workers are workers engaged for a limited time, specific project, seasonal need, or flexible business requirement. They may include temporary employees, contract workers, consultants, freelancers, and workers supplied by staffing agencies. In IT organizations, contingent workers are common in software testing, technical support, data annotation, cybersecurity audits, cloud migration, and ERP implementation.

5.1 Why Organizations Use Contingent Workers

- to quickly access specialized skills such as AI engineering, cloud security, DevOps, UI/UX, or compliance auditing;
- to handle short-term projects or deadlines;
- to reduce fixed employment costs;
- to manage uncertainty in demand;
- to obtain external expertise and fresh perspectives.

5.2 Ethical Risks

Risk	Explanation
Unequal treatment	Contingent workers may do similar work as permanent employees but receive fewer benefits, less recognition, or poorer working conditions.
Job insecurity	Short contracts may create stress and reduce bargaining power.
Hidden dependence	A worker may be called independent but economically depend on one company, creating a risk of disguised employment.
Training gaps	Temporary workers may be given access to sensitive systems without sufficient security or ethics training.
Confidentiality risk	Workers may handle trade secrets, source code, client data, or personal data without proper controls.
Access-control risk	Accounts may remain active after a contract ends, creating security exposure.

5.3 Responsible Management Practices

- Use clear written contracts defining scope, payment, confidentiality, ownership of work product, security duties, and dispute resolution.
- Provide basic onboarding on security, privacy, workplace conduct, and reporting channels.
- Apply least-privilege access and remove access immediately after the assignment ends.
- Avoid using contractor status to escape fair labour responsibilities.
- Give credit and feedback when contingent workers contribute significantly.
- Ensure that AI data workers receive clear labeling instructions, reasonable deadlines, and protection from harmful content exposure.

6. H-1B Workers

H-1B is a United States nonimmigrant worker classification used by employers to temporarily employ foreign workers in specialty occupations. USCIS describes H-1B specialty occupations as roles that require theoretical and practical application of a body of specialized knowledge. The U.S. Department of Labor explains that such occupations generally require at least a bachelor's degree or equivalent in a specific specialty.

This topic is included in IT ethics because many global technology companies employ skilled foreign professionals. Students should understand the ethical issues around skilled migration, dependency on employer sponsorship, wages, workplace fairness, and outsourcing/global labour markets.

6.1 Why H-1B Workers Are Important in IT

- Many IT jobs require specialized knowledge in software engineering, data science, cybersecurity, cloud computing, VLSI, AI/ML, and systems architecture.
- Global companies often operate across countries and recruit talent internationally.
- H-1B workers may contribute to innovation, research, product development, and technology transfer.
- The same system can raise ethical concerns if workers become dependent on one employer or if visa rules are misused.

6.2 Ethical Issues Related to H-1B Workers

Issue	Explanation
Dependency and power imbalance	A worker's immigration status may depend on the sponsoring employer. This can make it difficult to complain about unfair treatment.
Fair wages and working conditions	Employers must avoid exploiting workers by paying unfair wages or assigning unreasonable workloads.
Non-discrimination	Workers should not be discriminated against because of nationality, immigration status, accent, language background, or culture.
Honest job descriptions	Employers should not misrepresent job duties, work locations, salaries, or required skills.
Bench period and project pressure	Workers waiting for projects may face stress if they are not treated lawfully and fairly.
Cultural inclusion	Global teams should support respectful communication, anti-harassment policies, and inclusion.

7. Outsourcing

Outsourcing means giving certain work or business functions to an external organization or individual instead of doing everything internally. IT outsourcing may include software development, application maintenance, customer support, data entry, data annotation, payroll processing, cybersecurity monitoring, cloud management, testing, and infrastructure support.

7.1 Types of Outsourcing in IT

Type	Meaning
Onshore outsourcing	Vendor is in the same country.
Nearshore outsourcing	Vendor is in a nearby country or similar time zone.
Offshore outsourcing	Vendor is in a distant country, often to reduce cost or access skills.
Business Process Outsourcing (BPO)	External handling of business processes such as support, payroll, or claims processing.
IT Outsourcing (ITO)	External handling of software, infrastructure, cloud, or cybersecurity operations.
Knowledge Process Outsourcing (KPO)	External handling of high-skill knowledge tasks such as analytics, research, or legal process support.
Cloud/SaaS outsourcing	Use of external cloud platforms, APIs, or software services.

7.2 Benefits of Outsourcing

- access to specialized skills and global talent;
- cost flexibility and faster scaling;
- ability to focus on core business functions;
- 24/7 support through distributed teams;
- faster product development when managed responsibly.

7.3 Ethical and Legal Risks in Outsourcing

Risk	Explanation
Data protection risk	Vendors may process personal data, trade secrets, or confidential customer information. Contracts must define security, privacy, retention, and breach-reporting duties.
Quality risk	Poorly managed outsourcing can reduce product quality and user safety.
Labour exploitation risk	Cost reduction should not depend on poor wages, unsafe conditions, excessive hours, or lack of worker protections.
Loss of accountability	The client organization remains ethically responsible even if work is performed by a vendor.
Cyber supply-chain risk	Vendors may introduce vulnerabilities, insecure software components, or weak access controls.
Cultural and communication issues	Misunderstanding requirements can cause errors, rework, and conflict.
Dependency and lock-in	The organization may become dependent on one vendor, making it difficult to change providers later.

7.4 Ethical Outsourcing Checklist

- Choose vendors using transparent criteria, not bribes, favouritism, or hidden conflicts of interest.
- Check vendor policies on security, privacy, labour conditions, environmental responsibility, and anti-corruption.
- Use clear contracts, service-level agreements, audit rights, data-processing terms, and confidentiality clauses.
- Apply cybersecurity supply-chain risk management, including vendor risk assessment and continuous monitoring.
- Do not outsource work in a way that hides unethical practices from customers or regulators.
- Protect affected employees through reskilling, fair communication, and transition support where possible.

8. Whistleblowing

Whistleblowing means reporting serious wrongdoing inside an organization or by an organization. In IT, wrongdoing may include data theft, privacy violations, security cover-ups, false claims about AI performance, unsafe software release, and corruption in procurement, harassment, discrimination, environmental violations, or illegal surveillance.

8.1 Types of Whistleblowing

Type	Meaning
Internal whistleblowing	Reporting wrongdoing through internal channels such as supervisor, ethics committee, compliance officer, audit committee, HR, or secure reporting portal.
External whistleblowing	Reporting to external authorities, regulators, law enforcement, media, or public-interest bodies, usually when internal channels fail or immediate public harm exists.
Anonymous reporting	Reporting without revealing identity. It may protect the reporter but can make investigation more difficult.
Confidential reporting	Identity is known to a limited authorized team but protected from unnecessary disclosure.

8.2 When Can Whistleblowing Be Ethically Justified?

- The issue involves serious harm, illegality, fraud, corruption, public safety, privacy breach, discrimination, or major unethical conduct.
- The reporter has reasonable evidence or a good-faith belief, not merely personal dislike.
- Internal reporting channels have been tried or are unsafe, ineffective, or controlled by wrongdoers.
- The report is made to an appropriate authority or channel, not to cause unnecessary harm.
- The reporter avoids exposing unrelated confidential data.

8.3 Whistleblowing in Indian Context

India has the Whistle Blowers Protection Act, 2014, which was enacted to provide a mechanism for receiving complaints relating to corruption, wilful misuse of power or discretion, and related wrongdoing by public servants, and to provide safeguards against victimization [R7]. For companies, Section 177 of the Companies Act, 2013 and related rules require specified companies, including listed companies, to establish a vigil mechanism for directors and employees. The mechanism should provide safeguards against victimization and access to the audit committee chairperson or nominated director in appropriate cases [R8].

8.4 Responsibilities of an IT Organization

- Maintain confidential and accessible reporting channels.
- Protect whistleblowers from retaliation such as dismissal, demotion, harassment, salary reduction, threats, or project exclusion.
- Investigate complaints fairly and promptly.
- Protect evidence such as logs, emails, tickets, access records, model training data, and audit trails.
- Take corrective action if wrongdoing is confirmed.
- Protect falsely accused persons through fair process and confidentiality.

9. Green Computing

Green computing means designing, manufacturing, using, and disposing of computers, networks, data centres, software, and digital services in an environmentally responsible way. It aims to reduce energy use, carbon emissions, water use, toxic materials, and e-waste while maintaining useful computing services.

9.1 Why Green Computing Matters in IT Organizations

- IT equipment consumes electricity and often requires cooling, especially in data centres.
- Short device life cycles create e-waste containing valuable materials and hazardous substances.
- AI/ML workloads may require large amounts of compute for training and inference.
- Cloud services hide physical infrastructure from users, but energy and hardware impacts still exist.
- Sustainable IT can reduce cost, improve reputation, and support environmental responsibility.

9.2 Green Computing Practices

Practice	Explanation
Energy-efficient hardware	Use efficient processors, power supplies, monitors, and cooling systems.
Virtualization and cloud optimization	Run workloads efficiently and avoid underused servers.
Efficient software design	Write code that uses less CPU, memory, storage, and network bandwidth.
AI model efficiency	Use smaller models, pruning, quantization, distillation, caching, batching, and efficient inference.
Device life extension	Repair, upgrade, reuse, and donate equipment when safe and lawful.
Responsible procurement	Choose products with energy-efficiency and repairability considerations.
E-waste recycling	Use authorized recyclers and follow legal e-waste management requirements.
Environmental management systems	Use systematic planning, measurement, auditing, and improvement, such as ISO 14001 environmental management practices.

9.3 E-Waste and Indian Regulatory Context

Electronic waste includes end-of-life computers, mobile phones, servers, networking equipment, storage devices, printers, and other electronic products. India's E-Waste (Management) Rules, 2022 were notified by the Ministry of Environment, Forest and Climate Change and are implemented through mechanisms including Extended Producer Responsibility (EPR), environmentally sound recycling, and compliance responsibilities for relevant entities.

For IT organizations, green computing is not only about switching off monitors. It includes responsible purchase, secure data erasure before disposal, reuse where safe, disposal through authorized channels, and documentation for compliance.

9.4 Green Computing in AI & ML Projects

- Estimate compute cost before training very large models.
- Use transfer learning instead of training from scratch when suitable.
- Choose model size according to the use case; bigger is not always better.
- Use efficient data pipelines to avoid repeated preprocessing.
- Monitor GPU/TPU utilization and avoid idle cloud resources.
- Prefer model compression for deployment on edge devices when accuracy remains acceptable.
- Report model limitations and resource use honestly in research or product documentation.

10. ICT Industry Code of Conduct

A code of conduct is a formal statement of expected behaviour. It tells employees, managers, suppliers, contractors, and sometimes users what standards the organization follows. In the ICT industry, codes of conduct usually cover honesty, privacy, security, anti-corruption, anti-harassment, and respect for intellectual property, conflict of interest, fair labour, environmental responsibility, and responsible use of technology.

10.1 Why Codes of Conduct Are Important

- They convert broad ethical values into practical workplace rules.
- They guide employees when law or policy is unclear.
- They create a standard for disciplinary action and reporting.
- They support trust with clients, users, regulators, and investors.
- They help contractors and suppliers understand required behaviour.
- They provide a base for training, audits, and compliance checks.

10.2 Major Professional and Industry References

Reference	Relevance to ICT organizations
ACM Code of Ethics	A professional code for computing practitioners. It emphasizes contributing to society and human well-being, avoiding harm, honesty, fairness, privacy, security, professional competence, and public good.
IEEE ethics resources	IEEE ethics materials emphasize integrity, responsible behaviour, ethical professional conduct, and technology for humanity.
Responsible Business Alliance Code of Conduct	The RBA Code sets social, environmental, and ethical standards for global supply chains, including labour, health and safety, environment, ethics, and management systems.
UN Global Compact Ten Principles	Principles for responsible business in human rights, labor, environment, and anti-corruption.
OECD responsible business conduct and technology	Highlights that technology risks to human rights, workers, democratic governance, and the environment must be addressed.

10.3 Typical Contents of an ICT Organization Code of Conduct

- obeying applicable laws and regulations;
- honest communication with customers, investors, and regulators;
- protection of confidential information and trade secrets;
- privacy and data-protection obligations;
- secure use of systems, passwords, devices, and networks;
- respect for open-source licences and intellectual property;
- avoidance and disclosure of conflicts of interest;
- anti-bribery and anti-corruption rules;
- anti-harassment, diversity, and respectful workplace rules;
- fair treatment of employees, contractors, vendors, and users;
- safe and responsible AI/ML development;
- environmental responsibility and e-waste handling;
- whistleblowing and non-retaliation policies.

10.4 Code of Conduct for AI & ML Teams

- Use datasets lawfully and ethically; document data sources and consent/usage restrictions.
- Check models for bias, privacy leakage, unsafe outputs, and security vulnerabilities.
- Do not exaggerate model accuracy or hide known limitations.
- Use human oversight for high-stakes domains such as healthcare, finance, education, policing, and hiring.
- Protect model weights, training data, prompts, logs, and API keys.
- Maintain reproducible documentation, version control, evaluation reports, and incident logs.
- Respect affected people by providing explanations, complaint paths, and correction mechanisms where appropriate.

Disclaimer

These notes are AI-generated for educational purposes and may contain minor inaccuracies or differ slightly from official study material. Please verify important topics with your syllabus and classroom notes.

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